

YouTube Webinar Summary

Source: <https://www.youtube.com/watch?v=nShiALbl6D8>

Total Slides: 81

Table of Contents

- [Slide 1 \(at 2.0s\)](#)
- [Slide 2 \(at 3.0s\)](#)
- [Slide 3 \(at 8.0s\)](#)
- [Slide 4 \(at 16.0s\)](#)
- [Slide 5 \(at 17.0s\)](#)
- [Slide 6 \(at 18.0s\)](#)
- [Slide 7 \(at 19.0s\)](#)
- [Slide 8 \(at 20.0s\)](#)
- [Slide 9 \(at 21.0s\)](#)
- [Slide 10 \(at 22.0s\)](#)
- [Slide 11 \(at 23.0s\)](#)
- [Slide 12 \(at 26.0s\)](#)
- [Slide 13 \(at 27.0s\)](#)
- [Slide 14 \(at 28.0s\)](#)
- [Slide 15 \(at 29.0s\)](#)
- [Slide 16 \(at 30.0s\)](#)
- [Slide 17 \(at 31.0s\)](#)
- [Slide 18 \(at 32.0s\)](#)
- [Slide 19 \(at 33.0s\)](#)
- [Slide 20 \(at 34.0s\)](#)
- [Slide 21 \(at 35.0s\)](#)
- [Slide 22 \(at 36.0s\)](#)
- [Slide 23 \(at 37.0s\)](#)

- Slide 24 (at 38.0s)
- Slide 25 (at 39.0s)
- Slide 26 (at 40.0s)
- Slide 27 (at 92.0s)
- Slide 28 (at 162.0s)
- Slide 29 (at 267.0s)
- Slide 30 (at 320.0s)
- Slide 31 (at 468.0s)
- Slide 32 (at 477.0s)
- Slide 33 (at 578.0s)
- Slide 34 (at 641.0s)
- Slide 35 (at 727.0s)
- Slide 36 (at 741.0s)
- Slide 37 (at 806.0s)
- Slide 38 (at 1032.0s)
- Slide 39 (at 1100.0s)
- Slide 40 (at 1162.0s)
- Slide 41 (at 1256.0s)
- Slide 42 (at 1295.0s)
- Slide 43 (at 1368.0s)
- Slide 44 (at 1397.0s)
- Slide 45 (at 1398.0s)
- Slide 46 (at 1500.0s)
- Slide 47 (at 1578.0s)
- Slide 48 (at 1859.0s)
- Slide 49 (at 1878.0s)
- Slide 50 (at 1884.0s)
- Slide 51 (at 1890.0s)
- Slide 52 (at 1901.0s)
- Slide 53 (at 1902.0s)
- Slide 54 (at 1907.0s)

- Slide 55 (at 1932.0s)
- Slide 56 (at 1975.0s)
- Slide 57 (at 1976.0s)
- Slide 58 (at 1977.0s)
- Slide 59 (at 1978.0s)
- Slide 60 (at 1979.0s)
- Slide 61 (at 2031.0s)
- Slide 62 (at 2099.0s)
- Slide 63 (at 2104.0s)
- Slide 64 (at 2105.0s)
- Slide 65 (at 2106.0s)
- Slide 66 (at 2107.0s)
- Slide 67 (at 2108.0s)
- Slide 68 (at 2109.0s)
- Slide 69 (at 2110.0s)
- Slide 70 (at 2111.0s)
- Slide 71 (at 2112.0s)
- Slide 72 (at 2113.0s)
- Slide 73 (at 2114.0s)
- Slide 74 (at 2115.0s)
- Slide 75 (at 2116.0s)
- Slide 76 (at 2117.0s)
- Slide 77 (at 2118.0s)
- Slide 78 (at 2119.0s)
- Slide 79 (at 2120.0s)
- Slide 80 (at 2128.0s)
- Slide 81 (at 2130.0s)

Slide 1 (at 2.0s)



Timestamp: 2.0s



YouTube Link: <https://www.youtube.com/watch?t=2>



Slide Image



Transcript

Hi everybody, thank

Slide 2 (at 3.0s)



Timestamp: 3.0s



YouTube Link: <https://www.youtube.com/watch?t=3>



Slide Image



Transcript

So would you please raise your hand if you have any experience with VMware?
Alright,

Slide 3 (at 8.0s)



Timestamp: 8.0s



YouTube Link: <https://www.youtube.com/watch?t=8>



Slide Image



Transcript

Thanks a lot. Thanks a lot. So it gives us a good overview to see where we could maybe dive more into details and other

Slide 4 (at 16.0s)



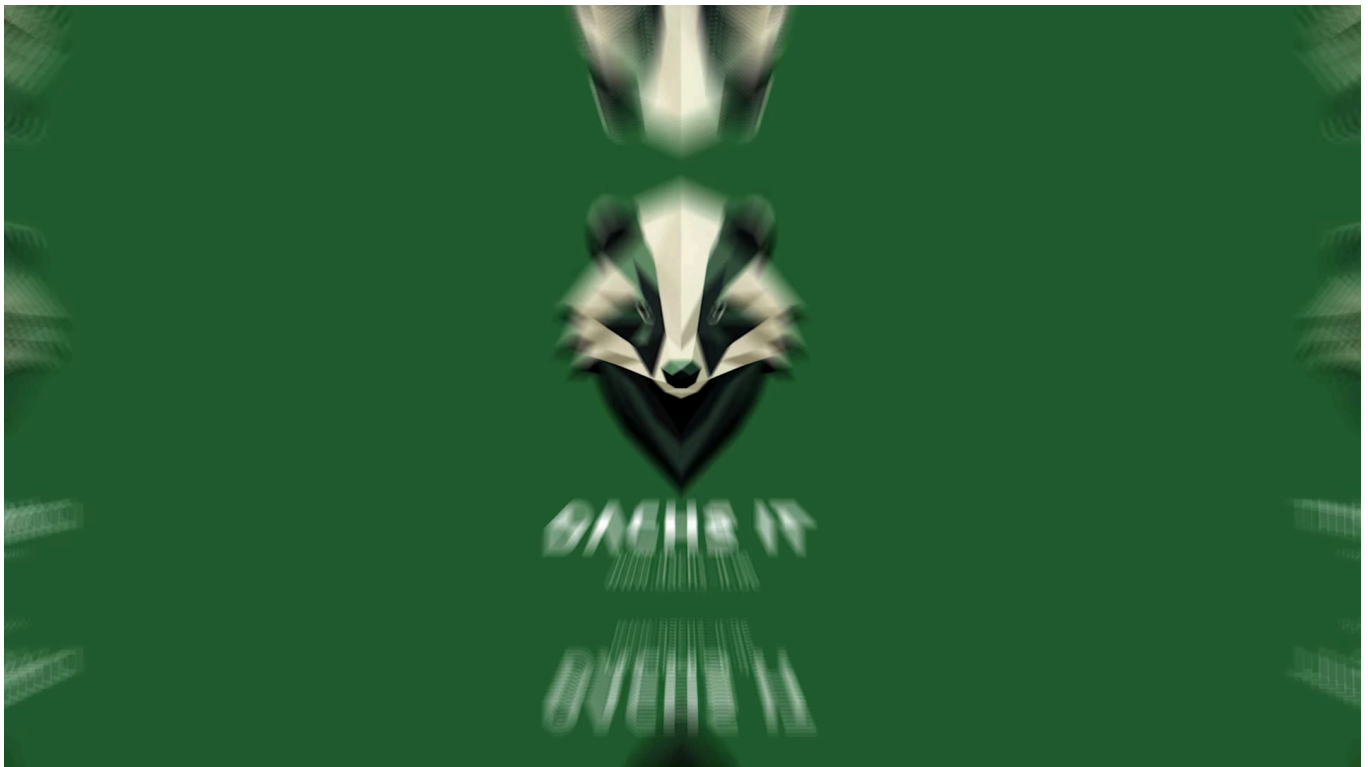
Timestamp: 16.0s



YouTube Link: <https://www.youtube.com/watch?t=16>



Slide Image



Transcript

almost like getting

Slide 5 (at 17.0s)

 **Timestamp:** 17.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=17>

Slide Image

From VMware to KubeVirt: A Technical Migration Journey

www.containerdays.io



ALEXANDER SCHABER
DACHS IT



RENE SCHRAMOWSKI
RED HAT



Transcript

reached a 25%

Slide 6 (at 18.0s)



Timestamp: 18.0s



YouTube Link: <https://www.youtube.com/watch?t=18>



Slide Image



Transcript

today. So I

Slide 7 (at 19.0s)

 **Timestamp:** 19.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=19>

Slide Image



Transcript

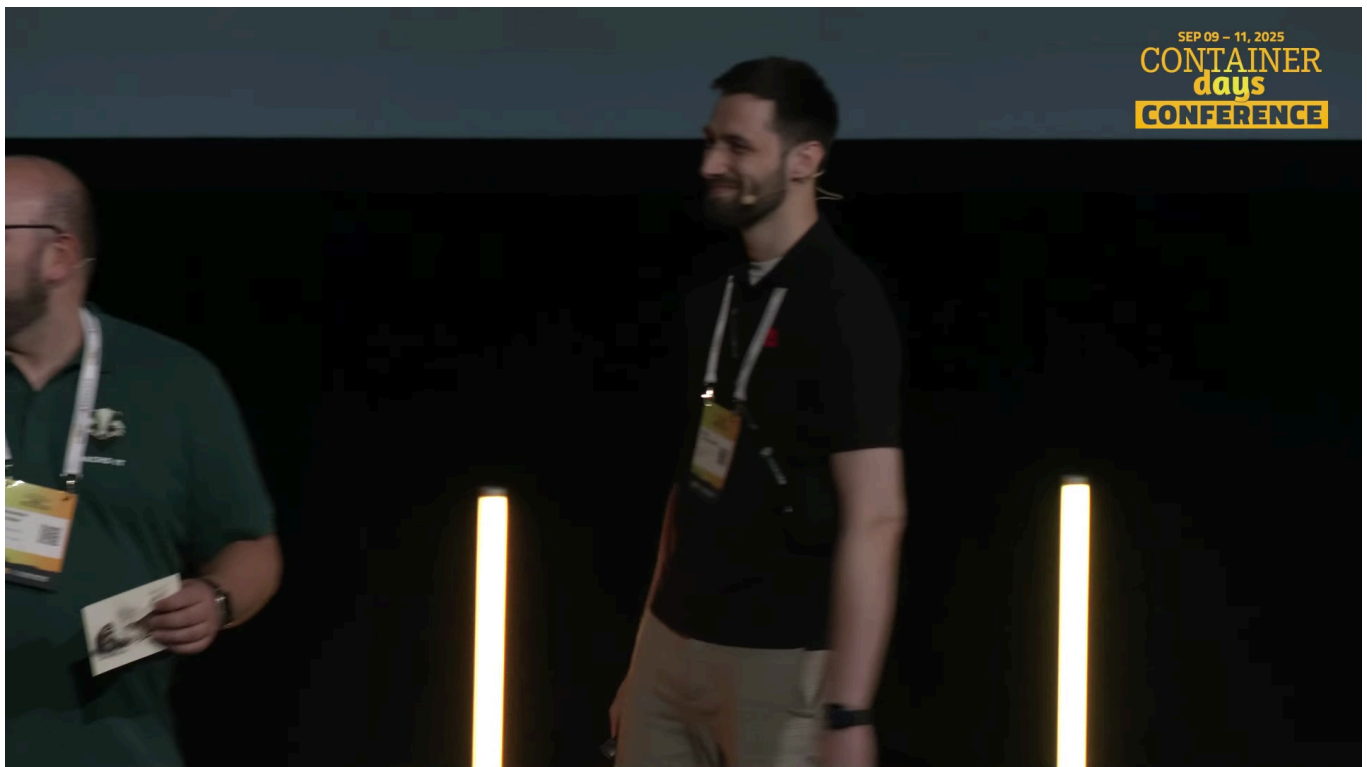
or teams working

Slide 8 (at 20.0s)

 **Timestamp:** 20.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=20>

Slide Image



Transcript

that. What we

Slide 9 (at 21.0s)

 **Timestamp:** 21.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=21>

Slide Image



Transcript

see here are

Slide 10 (at 22.0s)

 **Timestamp:** 22.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=22>

Slide Image



Transcript

is a CNCF

Slide 11 (at 23.0s)

 **Timestamp:** 23.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=23>

Slide Image



Transcript

awesome. Nice to say. I mean, it started, I

Slide 12 (at 26.0s)

 **Timestamp:** 26.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=26>

Slide Image



Transcript

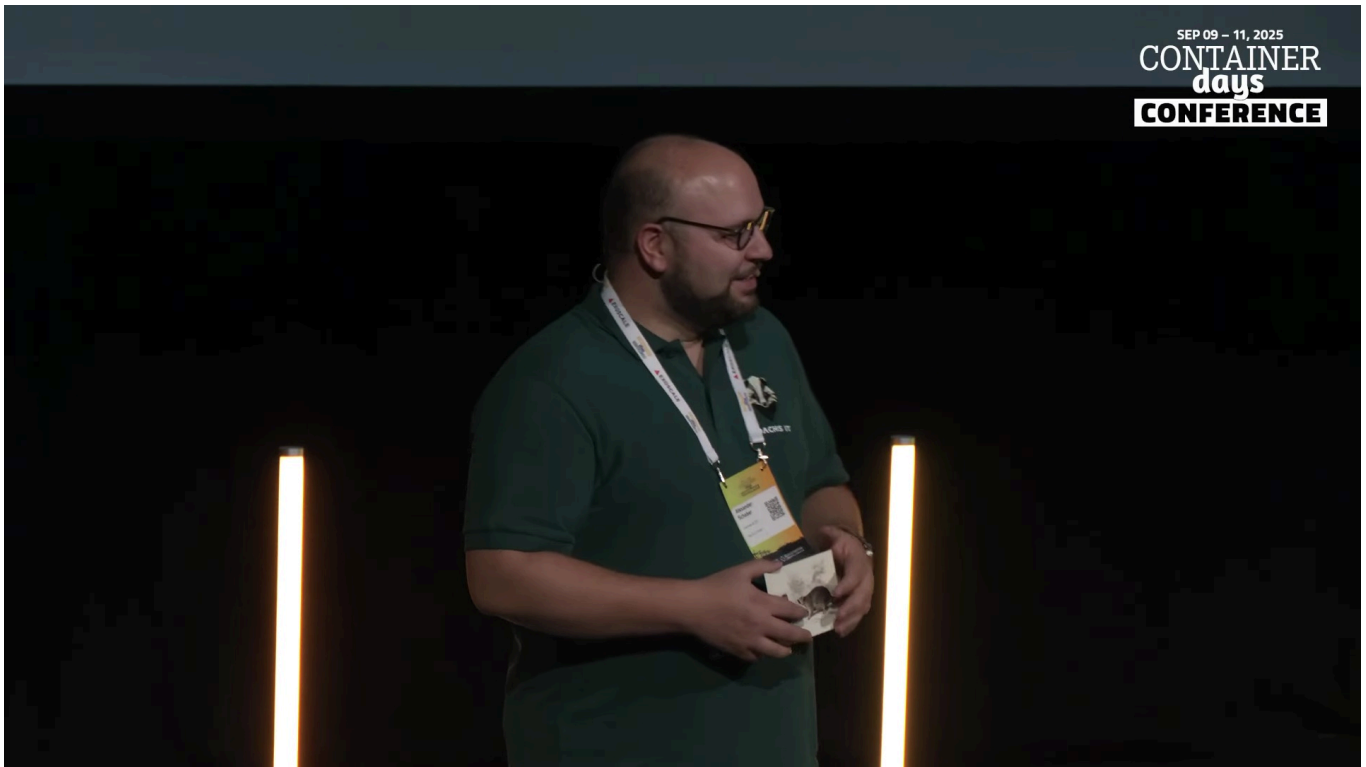
of you might

Slide 13 (at 27.0s)

 **Timestamp:** 27.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=27>

Slide Image



Transcript

So at the

Slide 14 (at 28.0s)

 **Timestamp:** 28.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=28>

Slide Image



Transcript

virtual machines. So

Slide 15 (at 29.0s)



Timestamp: 29.0s



YouTube Link: <https://www.youtube.com/watch?t=29>



Slide Image

SEP 09 - 11, 2025

CONTAINER
days
CONFERENCE



Transcript

you the Kubernetes

Slide 16 (at 30.0s)



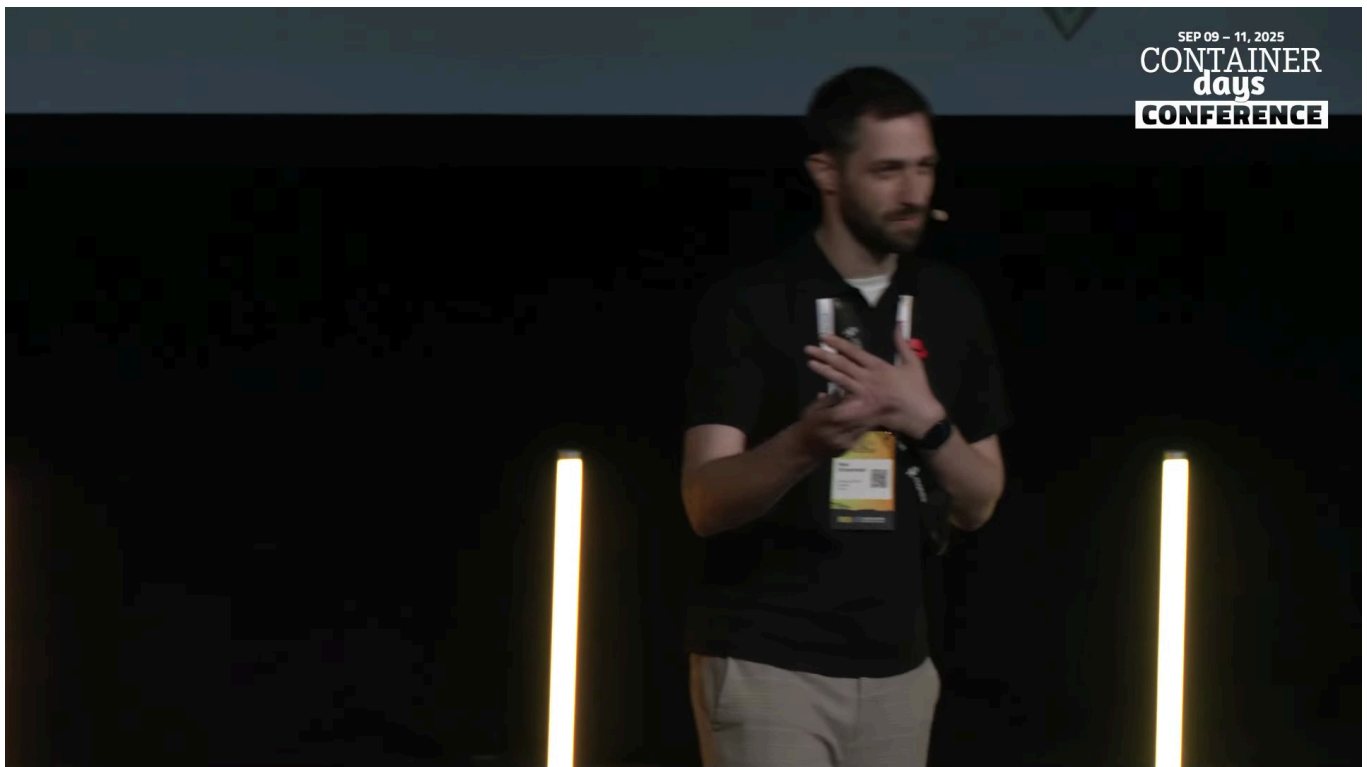
Timestamp: 30.0s



YouTube Link: <https://www.youtube.com/watch?t=30>



Slide Image



Transcript

you want to

Slide 17 (at 31.0s)



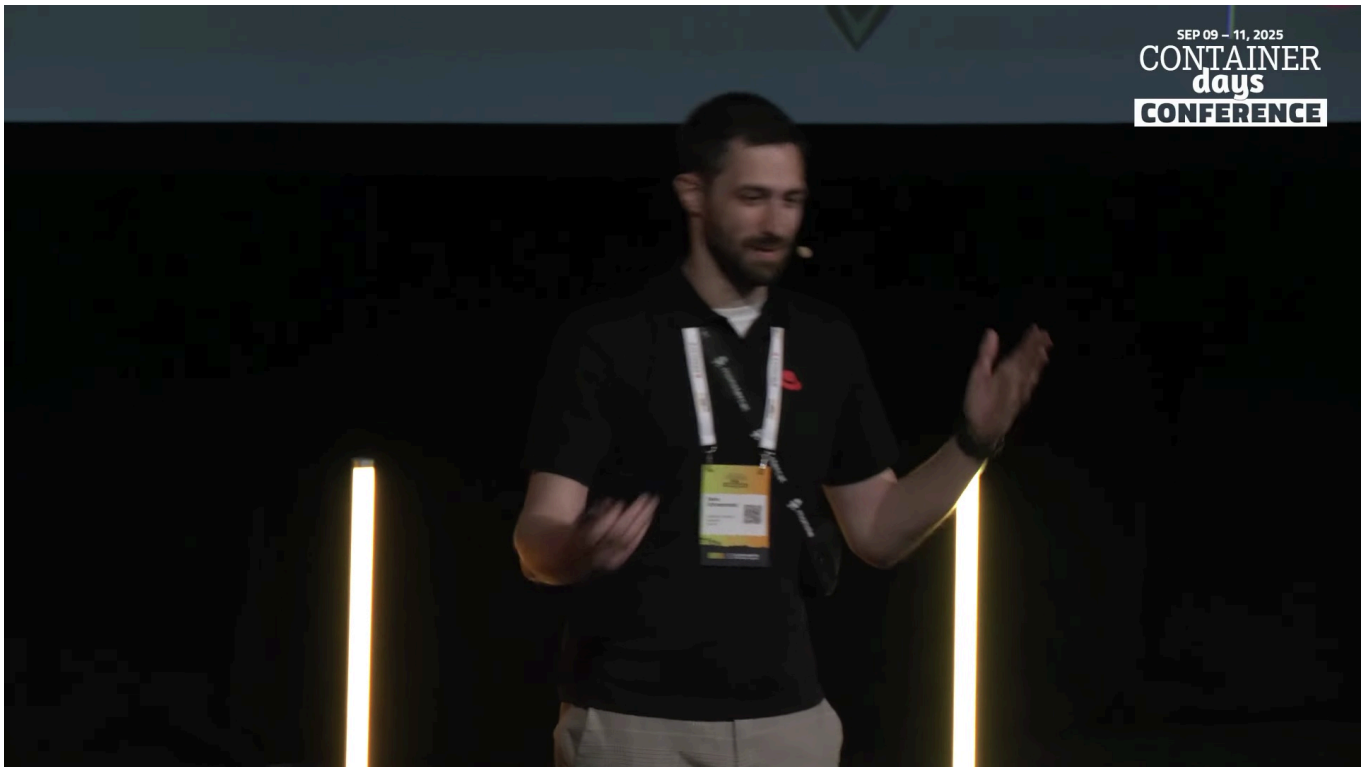
Timestamp: 31.0s



YouTube Link: <https://www.youtube.com/watch?t=31>



Slide Image



Transcript

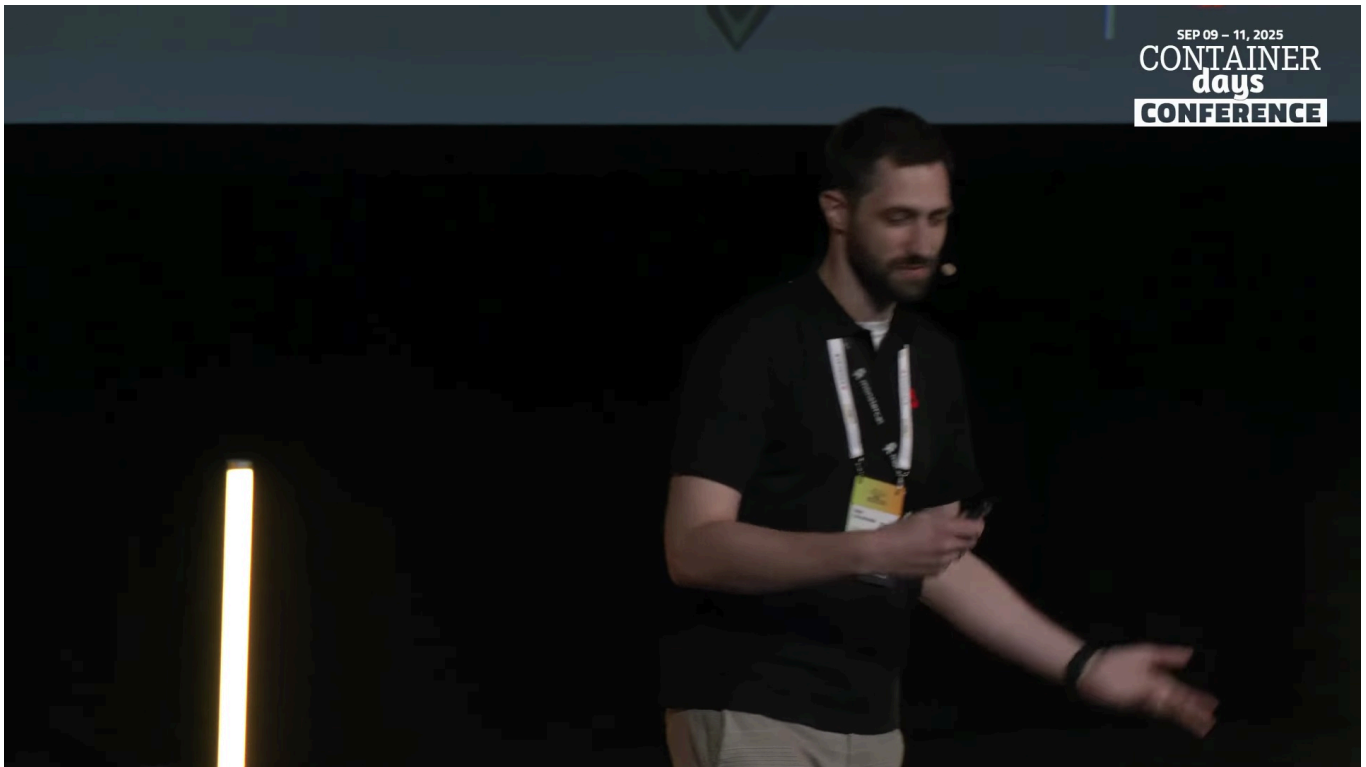
And I think,

Slide 18 (at 32.0s)

 **Timestamp:** 32.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=32>

Slide Image



Transcript

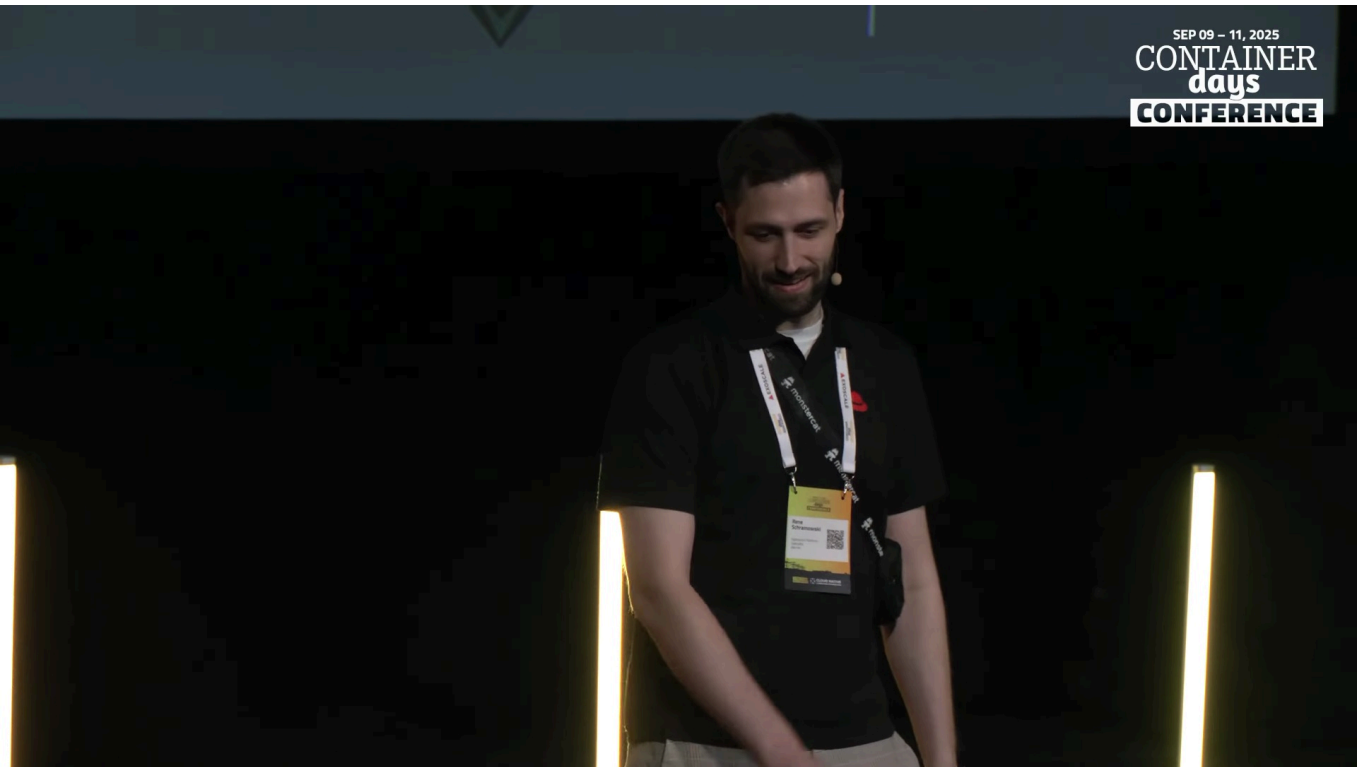
next to virtual

Slide 19 (at 33.0s)

 **Timestamp:** 33.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=33>

Slide Image



Transcript

container. There is

Slide 20 (at 34.0s)



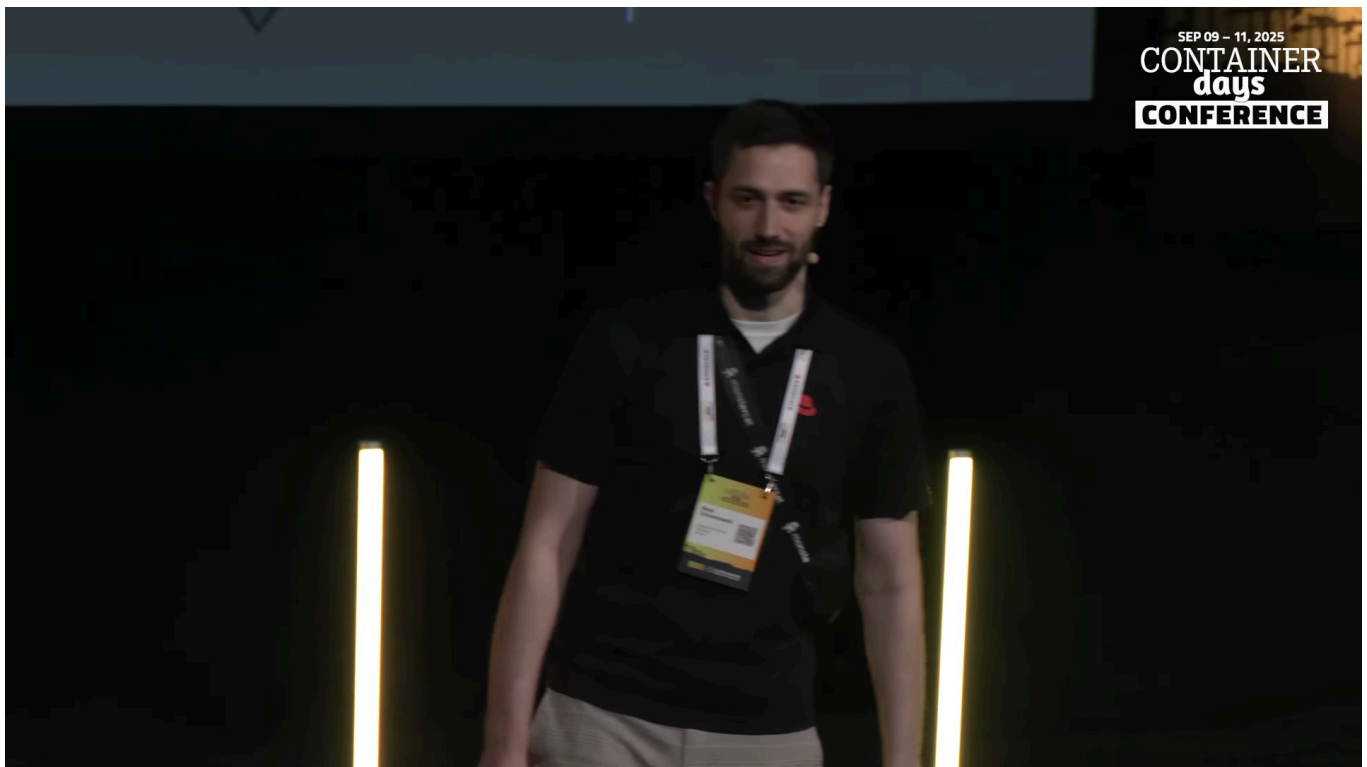
Timestamp: 34.0s



YouTube Link: <https://www.youtube.com/watch?t=34>



Slide Image



Transcript

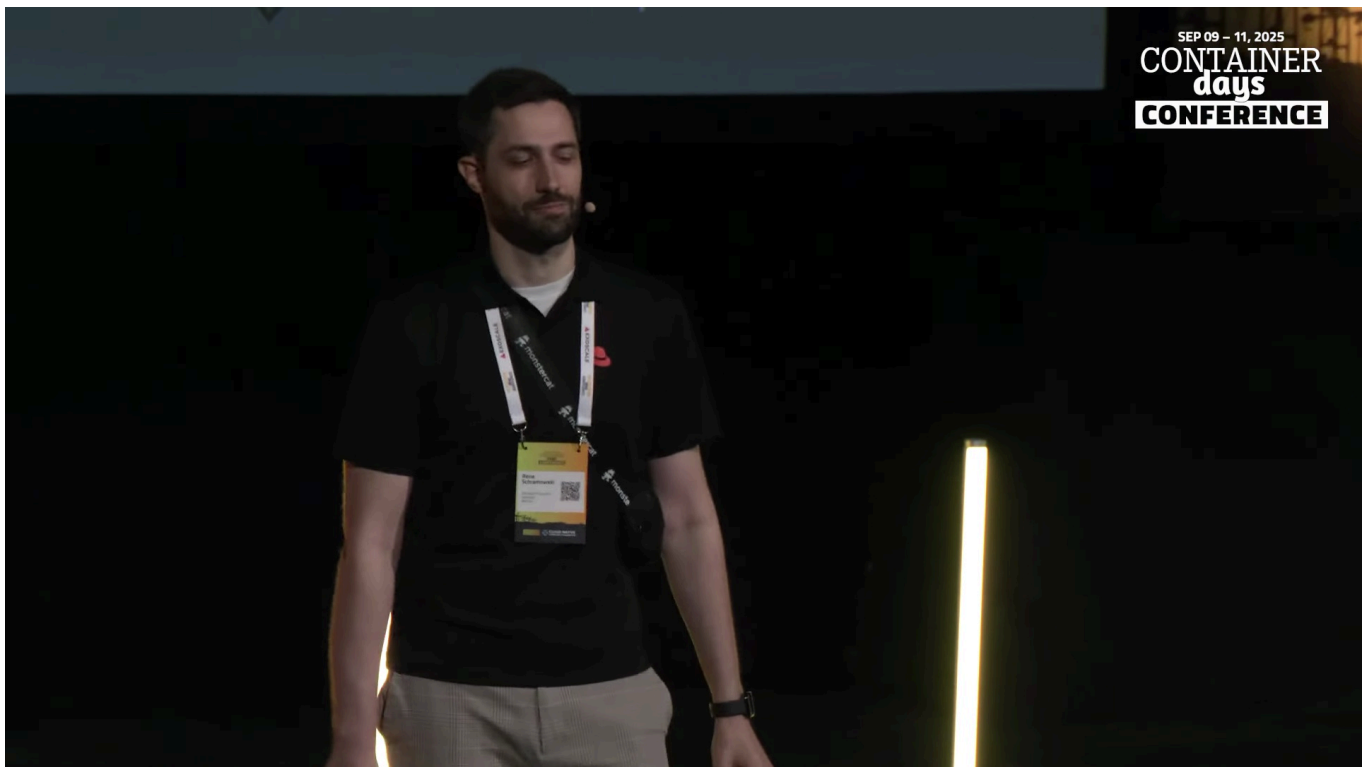
mean, that's what

Slide 21 (at 35.0s)

 **Timestamp:** 35.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=35>

Slide Image



Transcript

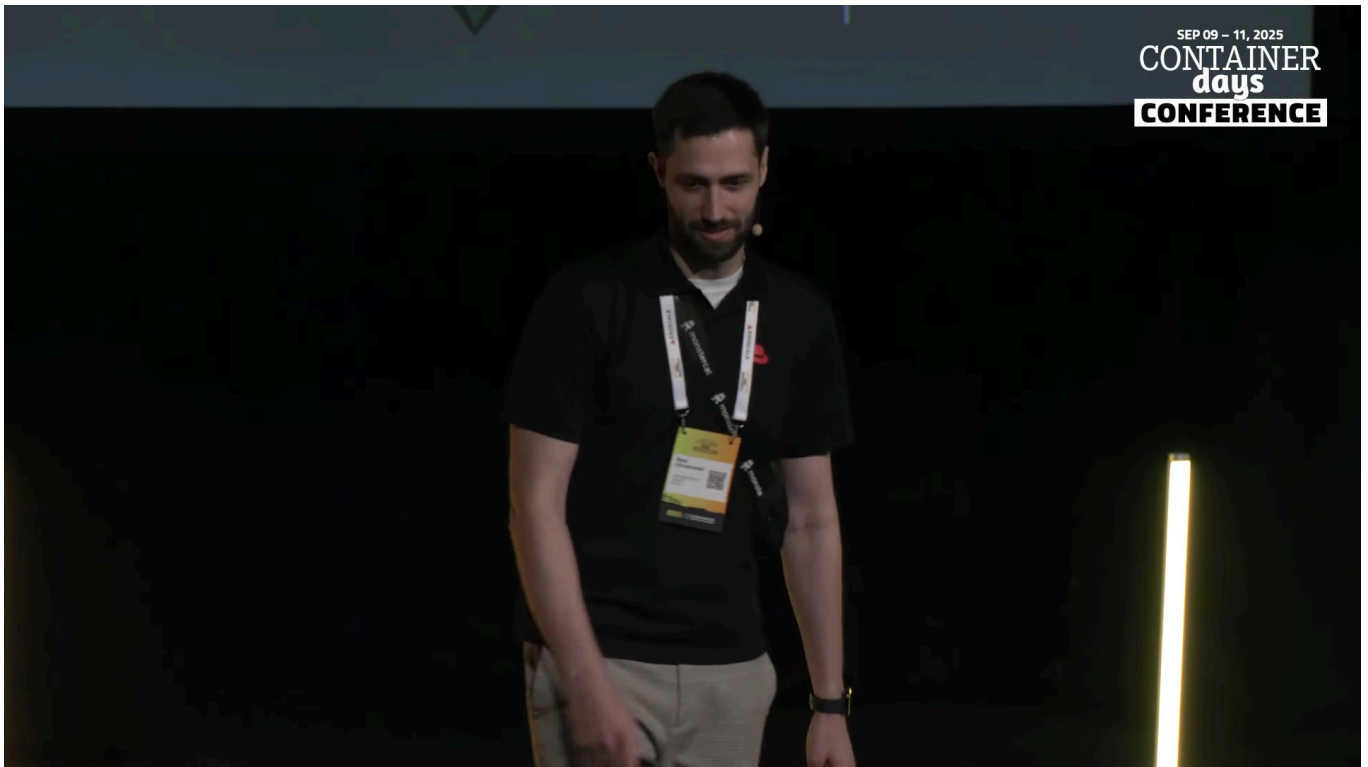
but fundamentally they

Slide 22 (at 36.0s)

 **Timestamp:** 36.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=36>

Slide Image



Transcript

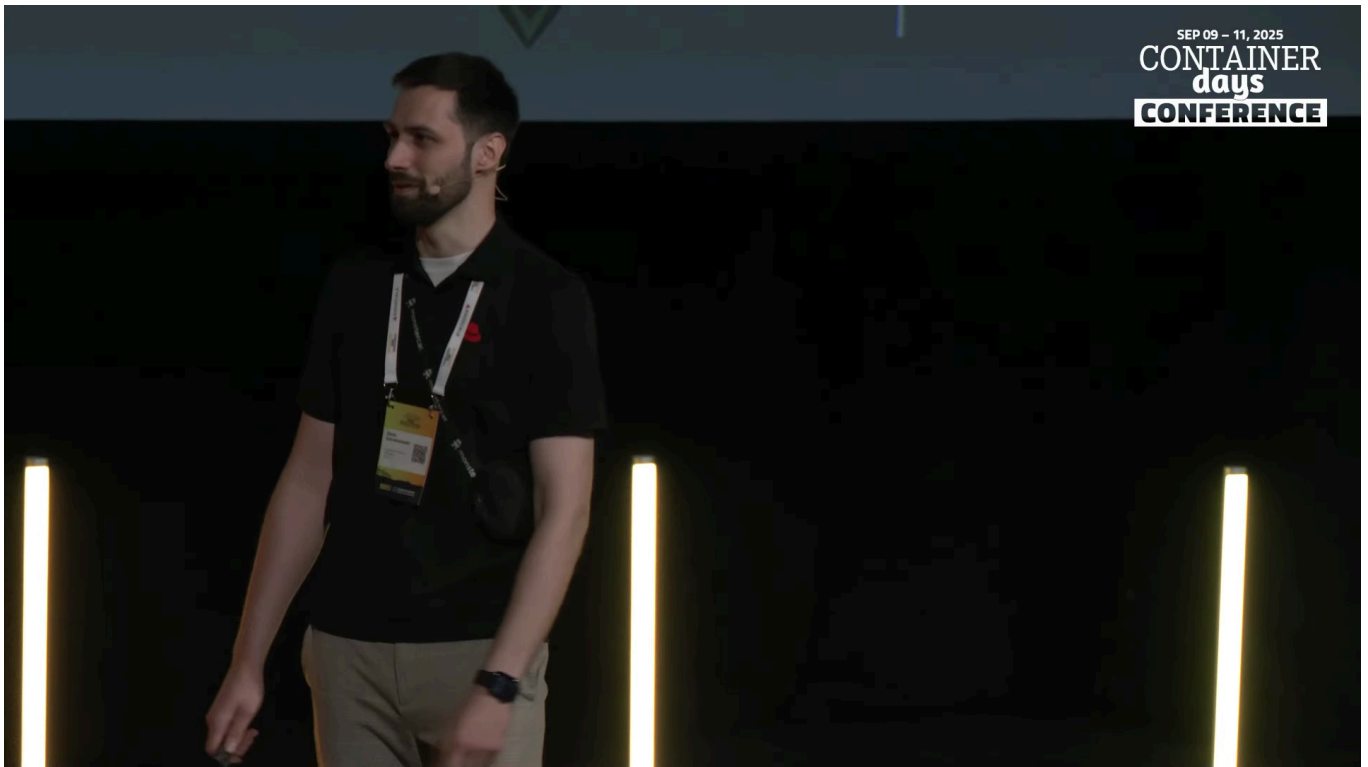
then everything else

Slide 23 (at 37.0s)

 **Timestamp:** 37.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=37>

Slide Image



Transcript

that's all it.

Slide 24 (at 38.0s)

 **Timestamp:** 38.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=38>

Slide Image



Transcript

a container. And

Slide 25 (at 39.0s)

 **Timestamp:** 39.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=39>

Slide Image



SEP 09 – 11, 2025
**CONTAINER
days
CONFERENCE**

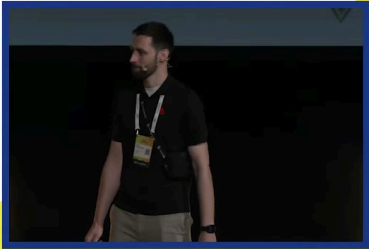
From VMware to KubeVirt: A Technical Migration Journey

**ALEXANDER SCHABER
AND RENE SCHRAMOWSKI**
DACHS IT & Red Hat

Alexander Schaber
Founder & CEO at DACHS IT
a.schaber@dachs-it.de

René Schramowski
Application Platform Specialist at Red Hat
rene@redhat.com



Transcript

this is a

Slide 26 (at 40.0s)



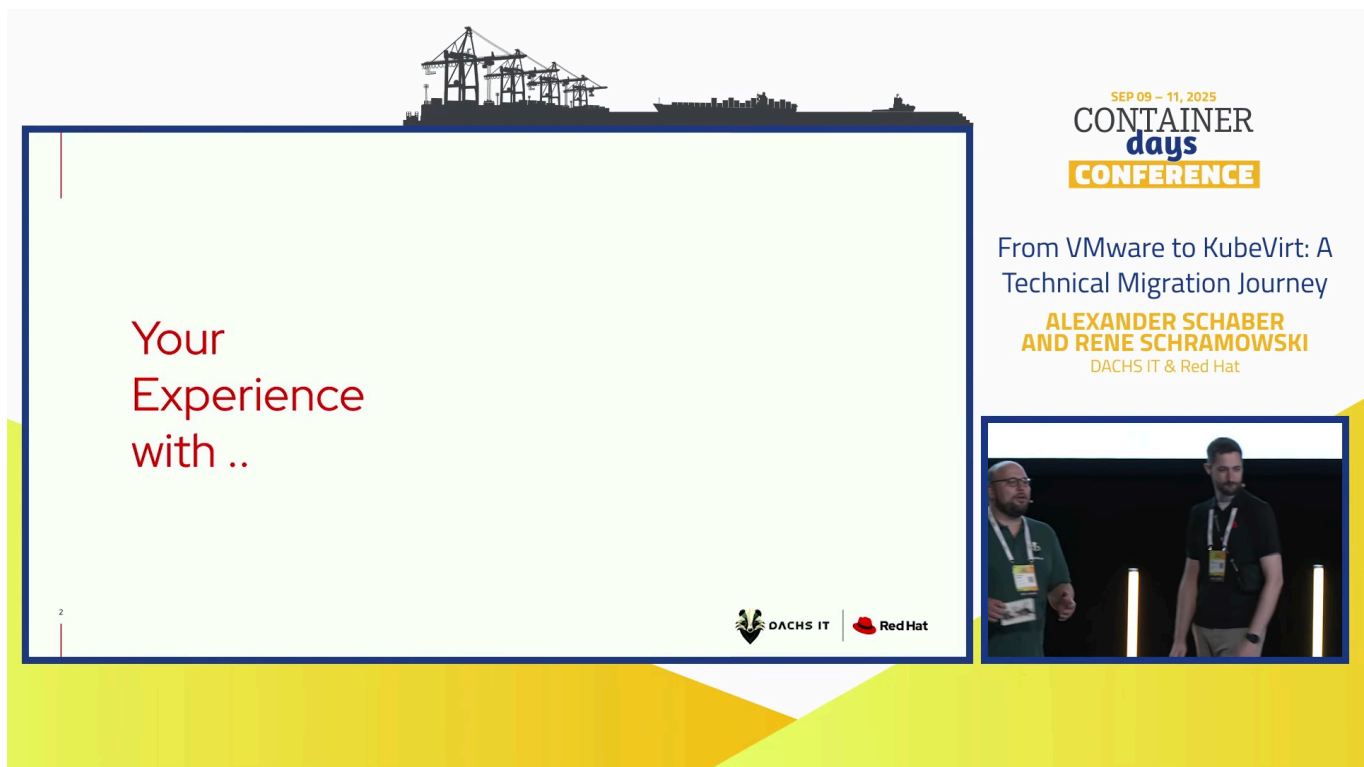
Timestamp: 40.0s



YouTube Link: <https://www.youtube.com/watch?t=40>



Slide Image



SEP 09 - 11, 2025
CONTAINER
days
CONFERENCE

From VMware to KubeVirt: A Technical Migration Journey

**ALEXANDER SCHABER
AND RENE SCHRAMOWSKI**
DACHS IT & Red Hat

Your
Experience
with ..

DACHS IT | Red Hat



Transcript

all pods. Correct. That's great. Great. Nice. Because I mean, this is one of the main challenges, right, because usually you have distinct teams managing virtual virtualization platforms and Kubernetes platform or containerization platforms, right. Usually distinct teams have in distinct knowledge and whatnot. And then now you're kind of trying to aggregate that and say, well, now I can, we can basically do both worlds on one platform. But storage, come on. Everybody needs storage. I mean, basically after compute storage is the next most important thing because you know, you want to have some storage where you can run your virtual machines off of. So can you maybe tell me a little bit about, you know, how does now storage work with containers and virtual machines inside them? Yes, definitely. Definitely. I mean, there are some caveats. It would have been too easy to just stop there and say, well, now you can have fun and just do it. There

Slide 27 (at 92.0s)



Timestamp: 92.0s



YouTube Link: <https://www.youtube.com/watch?t=92>

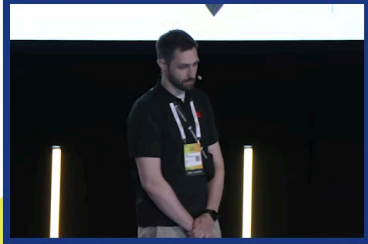
Slide Image



SEP 09 – 11, 2025
CONTAINER
days
CONFERENCE

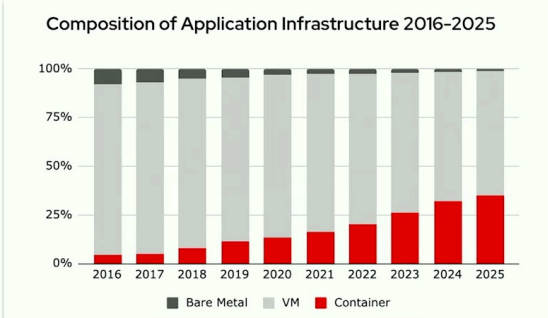
From VMware to KubeVirt: A Technical Migration Journey

**ALEXANDER SCHABER
AND RENE SCHRAMOWSKI**
DACHS IT & Red Hat



Industry Context for Virtualization Migration

Companies need to preserve their existing VM workloads while preparing for the future



Year	Bare Metal (%)	VM (%)	Container (%)
2016	10	85	5
2017	10	80	10
2018	10	75	15
2019	10	70	20
2020	10	65	25
2021	10	60	30
2022	10	55	35
2023	10	50	40
2024	10	45	45
2025	10	40	50

3
IDC Container Infrastructure Software Market Assessment: Container Deployment Forecast (IDC #US4867072)

 DACHS IT  Red Hat

Transcript

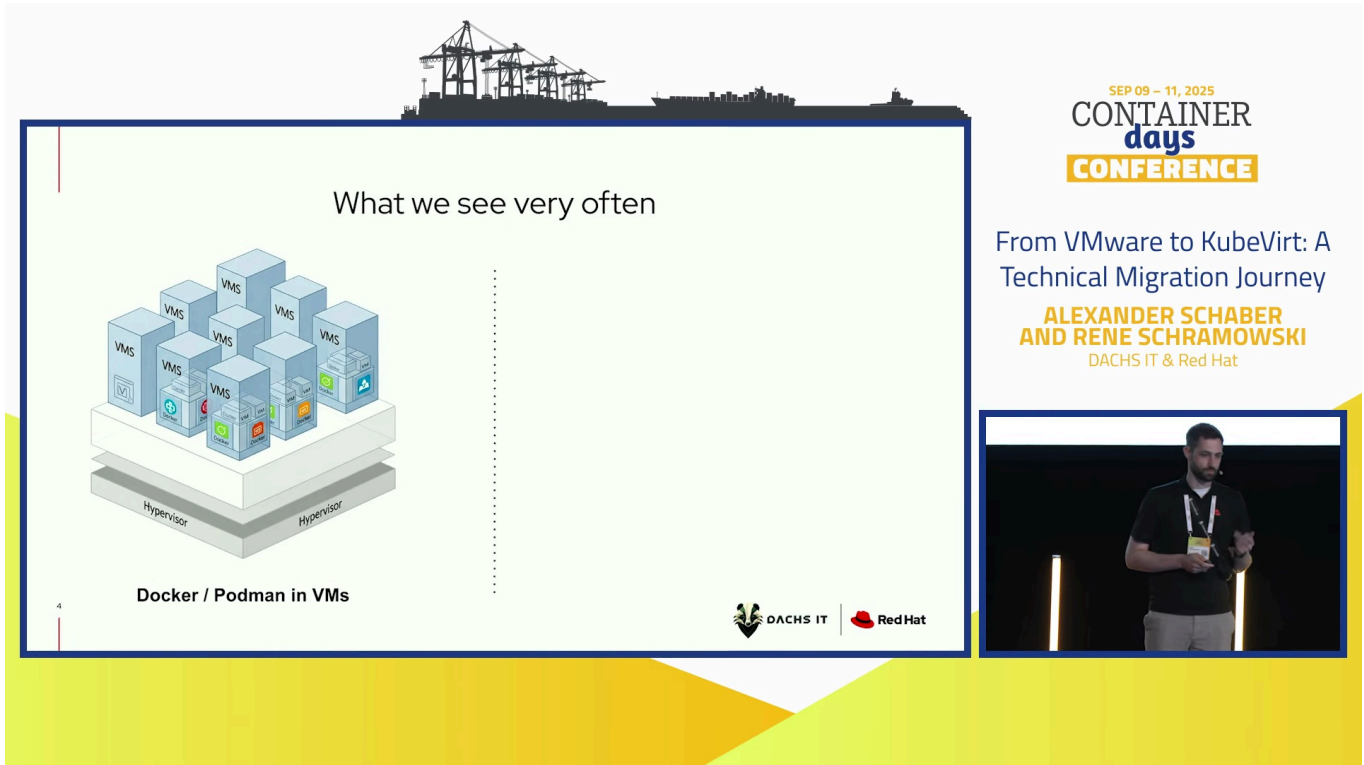
the next most important thing because you know, you want to have some storage where you can run your virtual machines off of. So can you maybe tell me a little bit about, you know, how does now storage work with containers and virtual machines inside them? Yes, definitely. Definitely. I mean, there are some caveats. It would have been too easy to just stop there and say, well, now you can have fun and just do it. There are some things you have to consider when doing that because even though I just said that both are, you also stated both are compute and CPU memory, whatnot. There's a lot of things they share still a virtual machine sometimes need a little bit more specific things. And the whole scope of things at the end of the day, since now the leading platform is Kubernetes, we now have to know that when we want to attach storage or use storage, we now have to go the Kubernetes way, right. So in Kubernetes, you know, there's the way of having your different CSI drivers for whatever kind of storage you want to attach. And then there's usually a way of either having storage classes or creating the PV PV so persistent volumes yourself. Then the PV is in

Slide 28 (at 162.0s)

 **Timestamp:** 162.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=162>

 **Slide Image**



What we see very often

Docker / Podman in VMs

SEP 09 – 11, 2025
**CONTAINER
days
CONFERENCE**

From VMware to KubeVirt: A
Technical Migration Journey

**ALEXANDER SCHABER
AND RENE SCHRAMOWSKI**
DACHS IT & Red Hat

DACHS IT | Red Hat

 **Transcript**

you have to consider when doing that because even though I just said that both are, you also stated both are compute and CPU memory, whatnot. There's a lot of things they share still a virtual machine sometimes need a little bit more specific things. And the whole scope of things at the end of the day, since now the leading platform is Kubernetes, we now have to know that when we want to attach storage or use storage, we now have to go the Kubernetes way, right. So in Kubernetes, you know, there's the way of having your different CSI drivers for whatever kind of storage you want to attach. And then there's usually a way of either having storage classes or creating the PV PV so persistent volumes yourself. Then the PV is in volume claims for whatever work to have and then kind of you said, right. So now we have to know that even though you're reusing virtual machines, we have to do is the Kubernetes way. But there are also some

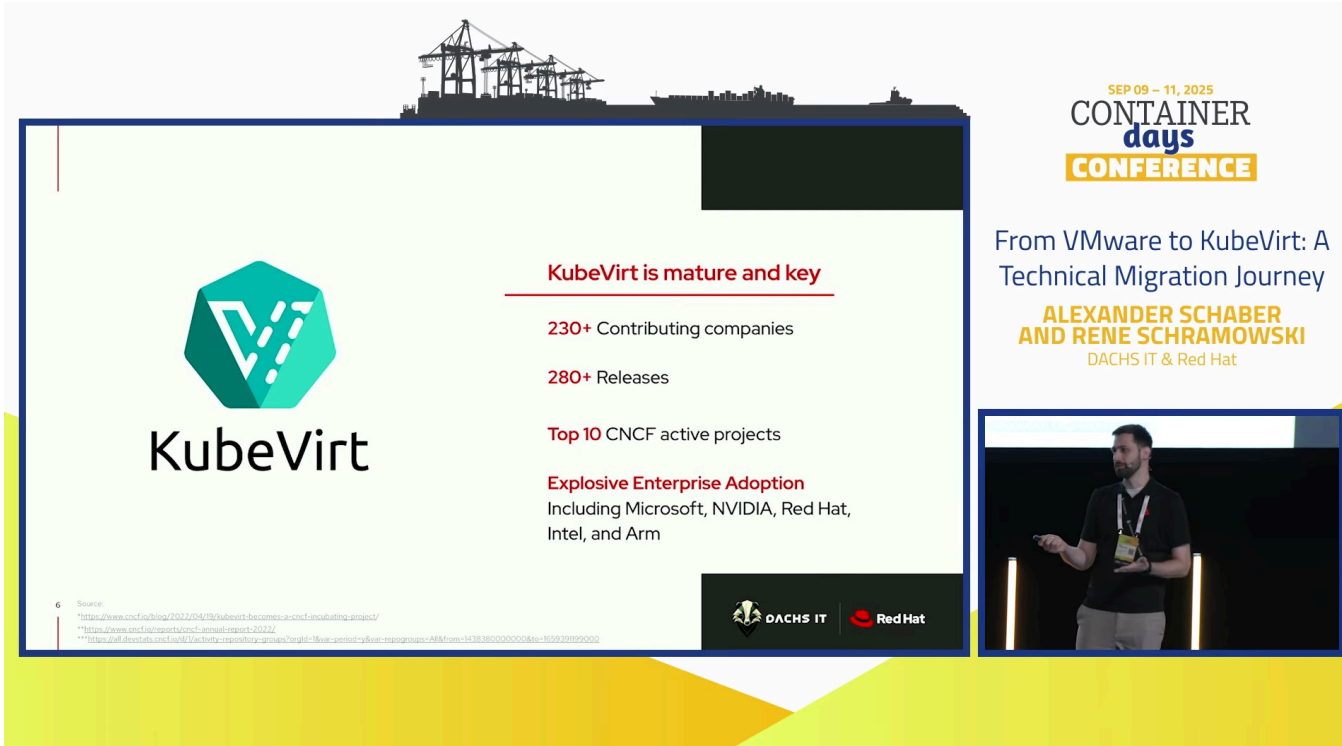
things we have to look at too. I mean, it's really cool that I already see a bunch of logos on the one hand side that show me that you don't need a new storage provider if you already have something that you've done with VMware. You can basically keep that storage and now make it Kubernetes compatible. I think, but I think there are a few things that I want to look into, right. I mean, maybe let's start. Yeah, I mean at the end of the day, one of the many things you would definitely like to leverage when talking about virtual machines or in general workload is that you are able to kind of use a live migration topic, right. Live migration and this scenario means that well, however big your cluster is that you want to kind of shift your workload from one node to another or whatever it

Slide 29 (at 267.0s)

 **Timestamp:** 267.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=267>

Slide Image



KubeVirt is mature and key

- 230+ Contributing companies
- 280+ Releases
- Top 10 CNCF active projects
- Explosive Enterprise Adoption**
Including Microsoft, NVIDIA, Red Hat, Intel, and Arm

SEP 09 – 11, 2025
CONTAINER days CONFERENCE

From VMware to KubeVirt: A Technical Migration Journey

ALEXANDER SCHABER AND RENE SCHRAMOWSKI
DACHS IT & Red Hat

6 Source:
*<https://www.cncf.io/blog/2022/04/19/kubevirt-becomes-a-cncf-incubating-project/>
**<https://www.cncf.io/reports/cncf-annual-report-2022/>
***<https://github.com/cncf/kubevirt-repository/issues/2041#issuecomment-1438380000&in=16293995000>

Transcript

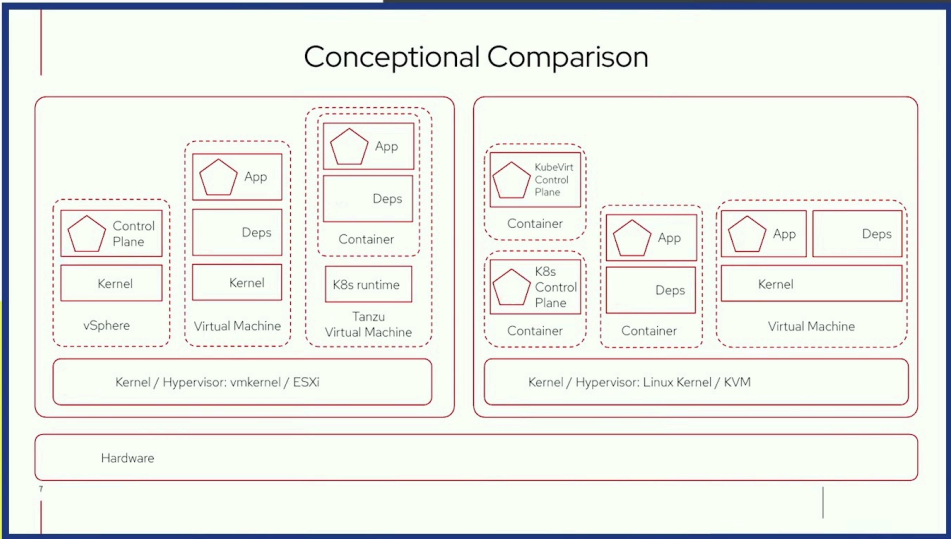
to go the Kubernetes way, right. So in Kubernetes, you know, there's the way of having your different CSI drivers for whatever kind of storage you want to attach. And then there's usually a way of either having storage classes or creating the PV so persistent volumes yourself. Then the PV is in volume claims for whatever work to have and then kind of you said, right. So now we have to know that even though you're reusing virtual machines, we have to do is the Kubernetes way. But there are also some things we have to look at too. I mean, it's really cool that I already see a bunch of logos on the one hand side that show me that you don't need a new storage provider if you already have something that you've done with VMware. You can basically keep that storage and now make it Kubernetes compatible. I think, but I think there are a few things that I want to

Slide 30 (at 320.0s)

 **Timestamp:** 320.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=320>

 **Slide Image**



The slide, titled "Conceptual Comparison", illustrates the architectural differences between VMware and Kubernetes. It is divided into two main sections: "vmkernel / ESXi" on the left and "Linux Kernel / KVM" on the right, both sitting on a "Hardware" base.

vmkernel / ESXi (Left):

- vSphere:** Contains a "Control Plane" and a "Kernel".
- Virtual Machine:** Contains an "App", "Deps", and "Kernel".
- Tanzu Virtual Machine:** Contains an "App", "Deps", "Container", and "K8s runtime".

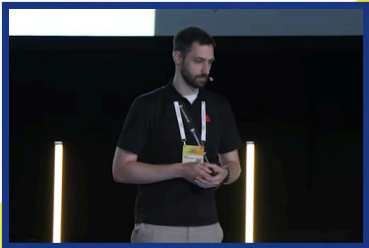
Linux Kernel / KVM (Right):

- Container:** Contains a "KubeVirt Control Plane" and a "K8s Control Plane".
- Container:** Contains an "App" and "Deps".
- Virtual Machine:** Contains an "App", "Deps", and "Kernel".

SEP 09 – 11, 2025
CONTAINER
days
CONFERENCE

From VMware to KubeVirt: A Technical Migration Journey

ALEXANDER SCHABER
AND RENE SCHRAMOWSKI
DACHS IT & Red Hat





Transcript

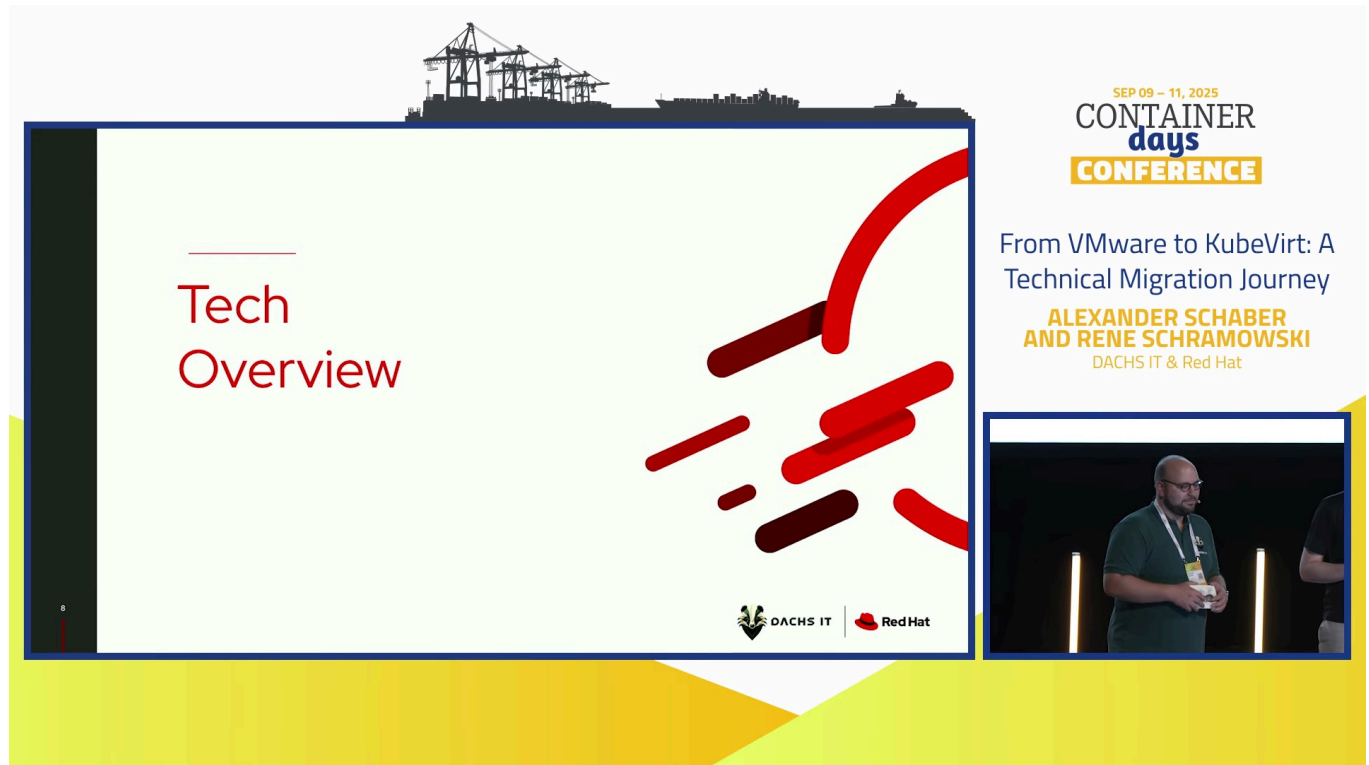
to do is the Kubernetes way. But there are also some things we have to look at too. I mean, it's really cool that I already see a bunch of logos on the one hand side that show me that you don't need a new storage provider if you already have something that you've done with VMware. You can basically keep that storage and now make it Kubernetes compatible. I think, but I think there are a few things that I want to look into, right. I mean, maybe let's start. Yeah, I mean at the end of the day, one of the many things you would definitely like to leverage when talking about virtual machines or in general workload is that you are able to kind of use a live migration topic, right. Live migration and this scenario means that well, however big your cluster is that you want to kind of shift your workload from one node to another or whatever it is, like shifting workload around. And you want to do the same with virtual machines and the thing is that if you want to do it with virtual machines via Qvert, your storage has to be able to do read right many. This is like one of them important things because we write many is nothing you kind of can do like take for granted sometimes you know a lot of times you work with read right months. And so for this scenario, you have to kind of think if you want to do live migration, you need to read right many. And also you have to kind of see like what are the other capabilities of my CSI driver because again coming from a VM world, you're usually using something like snapshotting or cloning or something like that. And if you want to do it in Kubernetes, US CSI driver has to have this capability and this is something you have to check because so if you just go ahead and you say like, oh, I heard that Qvert is awesome. Now I'm going to try it and don't really care about storage. This might be a neck breaker because this might be something where I say like when you really try it, you will see it. Okay, well, why doesn't it work and it is usually because of the capabilities of the CSI driver. That's kind of interesting. I mean, I remember when I tried to be motion for the very first time I had a server with direct attached storage and I'm like, okay, I want to do the motion and VM was like, nope, that's not how it works. So basically you have some requirements that you now also have for cube, which okay, kind of

Slide 31 (at 468.0s)

 **Timestamp:** 468.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=468>

Slide Image



Transcript

to look into, right. I mean, maybe let's start. Yeah, I mean at the end of the day, one of the many things you would definitely like to

Slide 32 (at 477.0s)

 **Timestamp:** 477.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=477>

Slide Image

SEP 09 – 11, 2025
CONTAINER
days
CONFERENCE

But Virtual Machine can be run INSIDE a container

- KVM virtual machine is a process
- Containers are meant to encapsulate processes
- Both have the same underlying resource needs
 - CPU, Memory, Storage, Network, etc
- Orchestration (K8s) layer is transparent for VM
- KVM has 10+ years of production use



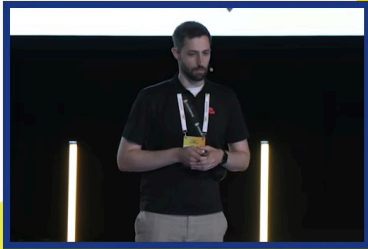
```

                    /usr/libexec/qemu-kvm -name
                    guest=wordpress-demo_wordpres
                    s-db,debug-threads=on -S -object
                    f'rom-type': 'secret'
                    ...
                    
```




From VMware to KubeVirt: A Technical Migration Journey

**ALEXANDER SCHABER
AND RENE SCHRAMOWSKI**
DACHS IT & Red Hat



Transcript

like shifting workload around. And you want to do the same with virtual machines and the thing is that if you want to do it with virtual machines via Qvert, your storage has to be able to do read right many. This is like one of them important things because we write many is nothing you kind of can do like take for granted sometimes you know a lot of times you work with read right months. And so for this scenario, you have to kind of think if you want to do live migration, you need to read right many. And also you have to kind of see like what are the other capabilities of my CSI driver because again coming from a VM world, you're usually using something like snapshotting or cloning or something like that. And if you want to do it in Kubernetes, US CSI driver has to have this capability and this is something you have to check because so if you just go ahead and you say like, oh, I heard that Qvert is awesome. Now I'm going to try it and don't really care about storage. This might be a neck breaker because this might be something where I say like when you really try it, you will see it. Okay, well, why doesn't it work and it is usually because of the capabilities of the CSI driver. That's kind of interesting. I mean, I remember when I tried to be motion for the very first time I had a server with direct attached storage and I'm like, okay, I want to do the motion and VM was like, nope, that's not how it works. So

basically you have some requirements that you now also have for cube, which okay, kind of makes sense. Another thing that you know, when I were on VMs is also usually very important because

Slide 33 (at 578.0s)



Timestamp: 578.0s



YouTube Link: <https://www.youtube.com/watch?t=578>



Slide Image



Virtual Machine can be run INSIDE a container

Pods > Pod details

virt-launcher-win2k19-tan-chipmunk-44-hxbgz Running

Details Metrics YAML Environment Logs Events **Terminal**

Connecting to **compute**

```
sh-5.1$ ps -aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
qemu      1  0.0  0.0 1679616 15624 ?        Ssl  13:54   0:00 /usr/bin/virt-launcher-monitor --qemu-timeout 342s --name win2k19-tan-chipmu
qemu     12  0.0  0.1 3517920 72824 ?        Sl   13:54   0:00 /usr/bin/virt-launcher --qemu-timeout 342s --name win2k19-tan-chipmunk-44 --
qemu     24  0.0  0.0 1303752 30020 ?        Sl   13:54   0:00 /usr/sbin/virtqemud -f /var/run/libvirt/virtqemud.conf
qemu     25  0.0  0.0 105272 15936 ?        Sl   13:54   0:00 /usr/sbin/virtlogd -f /etc/libvirt/virtlogd.conf
qemu     84  2.1  6.0 9554604 3908944 ?       Sl   13:54   0:10 /usr/libexec/qemu-kvm -name guest=rguske-demo_win2k19-tan-chipmunk-44,debug-
qemu    100  1.0  0.0  452 2456 pts/0    Ss   14:07   0:00 sh
qemu    115  0.0  0.0  15484 5760 pts/0    R+   14:07   0:00 ps -aux

sh-5.1$ uname -r
5.14.0-427.60.1.el9_4.x86_64
sh-5.1$ uname -a
Linux win2k19-tan-chipmunk-44 5.14.0-427.60.1.el9_4.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Mar 7 17:59:46 EST 2025 x86_64 x86_64 x86_64 GNU/Linux
sh-5.1$
```

10

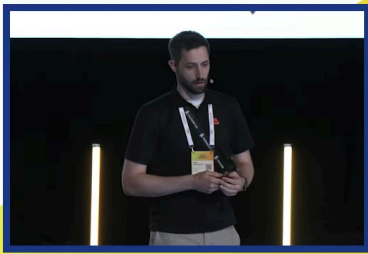
 DACHS IT  Red Hat

SEP 09 – 11, 2025

CONTAINER days CONFERENCE

From VMware to KubeVirt: A
Technical Migration Journey

**ALEXANDER SCHABER
AND RENE SCHRAMOWSKI**
DACHS IT & Red Hat





Transcript

you have to kind of think if you want to do live migration, you need to read right many. And also you have to kind of see like what are the other capabilities of my CSI driver because again coming from a VM world, you're usually using something like snapshotting or cloning or something like that. And if you want to do it in Kubernetes, US CSI driver has to have this capability and this is something you have to check because so if you just go ahead and you say like, oh, I heard that Qvert is awesome. Now I'm going to try it and don't really care

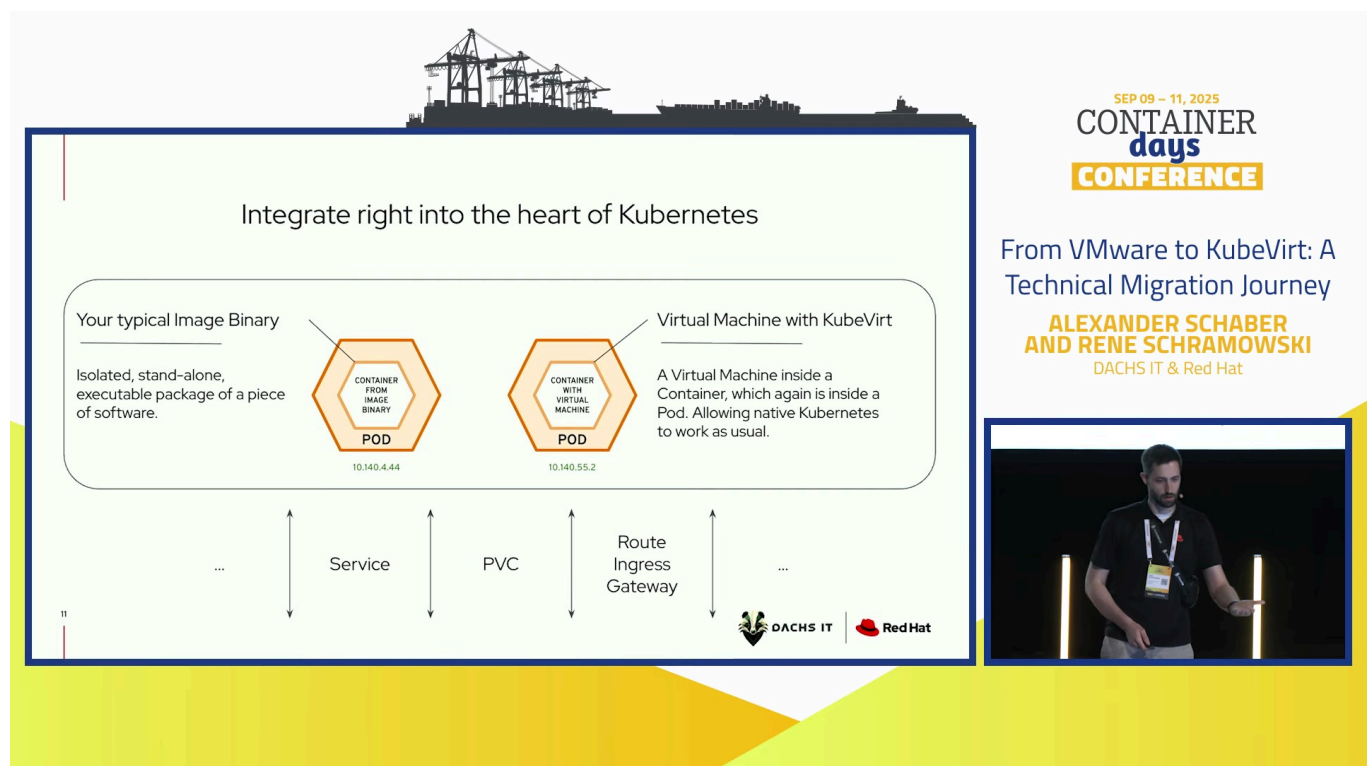
about storage. This might be a neck breaker because this might be something where I say like when you really try it, you will see it. Okay, well, why doesn't it work and it is usually because of the capabilities of the CSI driver. That's kind of interesting. I mean, I remember when I tried to be motion for the very first time I had a server with direct attached storage and I'm like, okay, I want to do the motion and VM was like, nope, that's not

Slide 34 (at 641.0s)

 **Timestamp:** 641.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=641>

 **Slide Image**



 **Transcript**

if you just go ahead and you say like, oh, I heard that Qvert is awesome. Now I'm going to try it and don't really care about storage. This might be a neck breaker because this might be something where I say like when you really try it, you will see it. Okay, well, why doesn't it work and it is usually because of the

capabilities of the CSI driver. That's kind of interesting. I mean, I remember when I tried to be motion for the very first time I had a server with direct attached storage and I'm like, okay, I want to do the motion and VM was like, nope, that's not how it works. So basically you have some requirements that you now also have for kube, which okay, kind of makes sense. Another thing that you know, when I were on VMs is also usually very important because VMs usually have more stateful applications. I think it's back up. So can you maybe tell us a little bit about backups. Sure, definitely definitely is also topic now again, Kubernetes is a leading platform. So now we have to think how do we do backup and restore and if everything works. And then how do we do it in Kubernetes, right? And again, you have to kind of think this is nothing out of the box in your best scenario. You already have something in place to tackle this topics. And there are, for example, either a lot of upstream open source tooling to do that or you have like a vendor of your choice who

Slide 35 (at 727.0s)



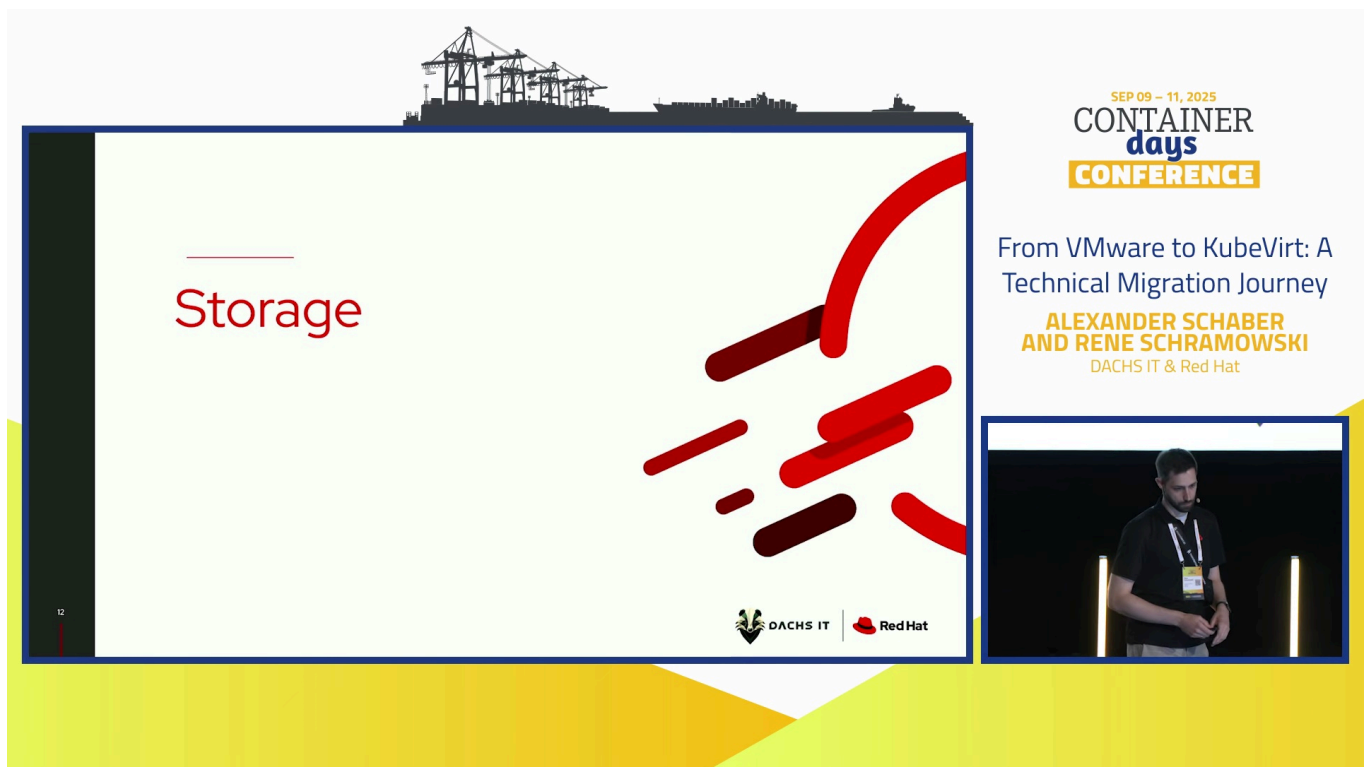
Timestamp: 727.0s



YouTube Link: <https://www.youtube.com/watch?t=727>



Slide Image



Transcript

be motion for the very first time I had a server with direct attached storage and I'm like, okay, I want to do the motion and VM was like, nope, that's not how it works. So basically you have some requirements that you

Slide 36 (at 741.0s)



Timestamp: 741.0s



YouTube Link: <https://www.youtube.com/watch?t=741>



Slide Image

Virtualization Storage

```

graph TD
    subgraph VMs_Containers [ ]
        direction LR
        VMs[VMs]
        Containers[Containers]
    end
    VMs_Containers --> SC[StorageClass]
    SC --> PVPV[PVC/PV]
    PVPV --> CSID[CSI Kubernetes Driver]
    CSID <--> VS[Vendor Storage  
(iSCSI, NFS, FC, etc.)]
    CSID --> PV[PV]
    VS --> LUN[LUN]
    PV -.->|Bind mount| VMs_Containers
    
```

Storage type: block, file, object
Performance tier: gold, silver, bronze

Request type: block, IOGB, read-write many

DACHS IT | Red Hat

SEP 09 – 11, 2025
**CONTAINER
days
CONFERENCE**

From VMware to KubeVirt: A
Technical Migration Journey

**ALEXANDER SCHABER
AND RENE SCHRAMOWSKI**
DACHS IT & Red Hat



Transcript

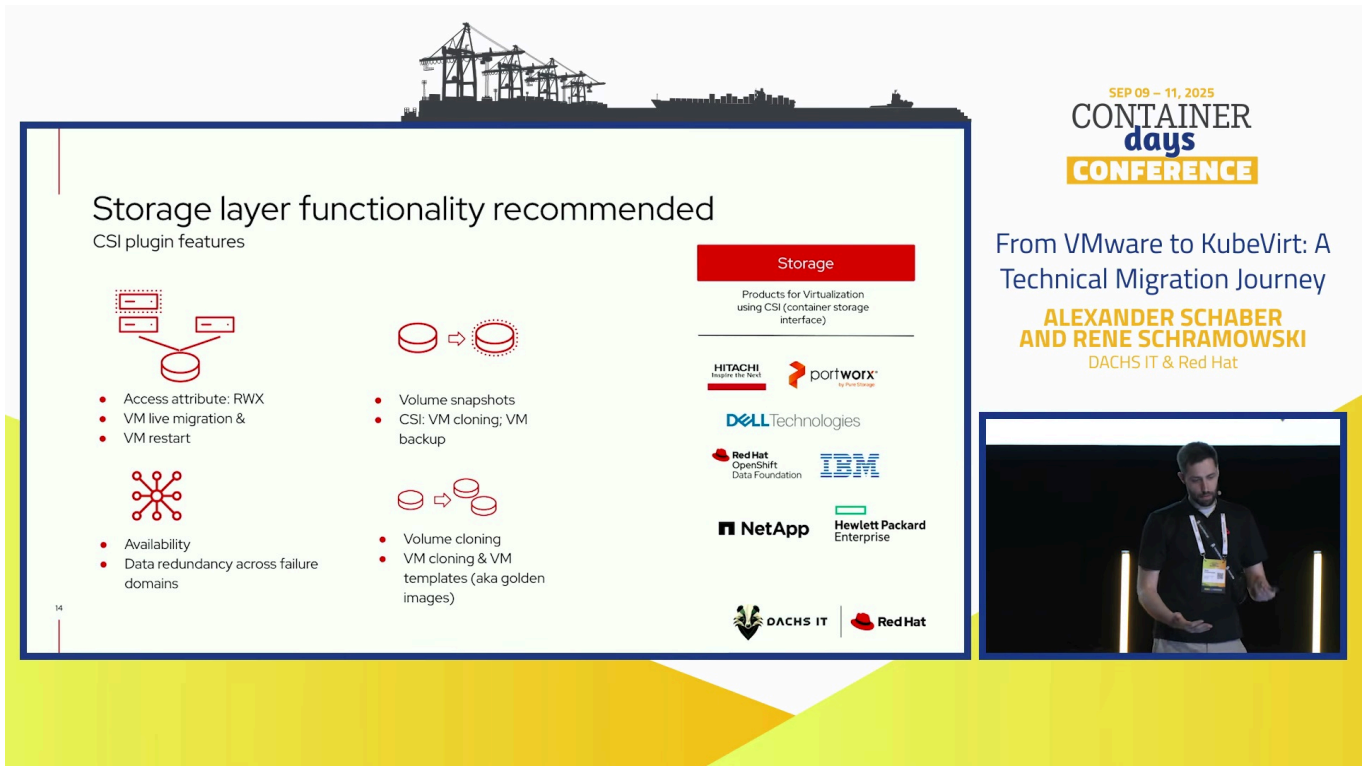
you maybe tell us a little bit about backups. Sure, definitely definitely is also topic now again, Kubernetes is a leading platform. So now we have to think how do we do backup and restore and if everything works. And then how do we do it in Kubernetes, right? And again, you have to kind of think this is nothing out of the box in your best scenario. You already have something in place to tackle this topics. And there are, for example, either a lot of upstream open source tooling to do that or you have like a vendor of your choice who already has a solution to do something like that on Kubernetes in general. Not naming any companies, but there are a lot of doing just that offering like backup recovery disaster recovery and helping you to do that on Kubernetes. And then you have to make sure that you know, your Kubernetes is ready for that because especially virtual machines where data is even more crucial and more baked into that is even more more important. And you can see just a very short one as I just mentioned all the CSI capabilities. You just have to check what is

Slide 37 (at 806.0s)

 **Timestamp:** 806.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=806>

 **Slide Image**



Storage layer functionality recommended
CSI plugin features

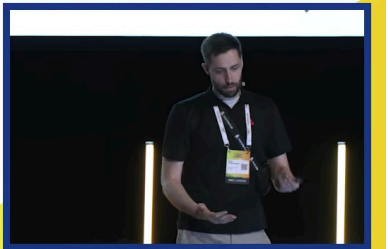
- Access attribute: RWX
- VM live migration & VM restart
- Volume snapshots
- CSI: VM cloning; VM backup
- Availability
- Data redundancy across failure domains
- Volume cloning
- VM cloning & VM templates (aka golden images)

Storage
Products for Virtualization using CSI (container storage interface)

HITACHI **portworx**
DELL Technologies
Red Hat OpenShift Data Foundation **IBM**
NetApp **Hewlett Packard Enterprise**
DACHS IT **Red Hat**

SEP 09 – 11, 2025
CONTAINER days CONFERENCE

From VMware to KubeVirt: A Technical Migration Journey
ALEXANDER SCHABER AND RENE SCHRAMOWSKI
DACHS IT & Red Hat



 **Transcript**

either a lot of upstream open source tooling to do that or you have like a vendor of your choice who already has a solution to do something like that on Kubernetes in general. Not naming any companies, but there are a lot of doing just that offering like backup recovery disaster recovery and helping you to do that on Kubernetes. And then you have to make sure that you know, your Kubernetes is ready for that because especially virtual machines where data is even more crucial and more baked into that is even more more important. And you can see just a very short one as I just mentioned all the CSI capabilities. You just have to check what is your vendor of choice for storage currently. And the best case is maybe some of the bigger vendors or namely vendors you know and they have well maintained CSI driver, you can just leverage. And in most cases, they are already ready for that. They already offer wheat white money. They already offer snapshotting and what so not. That's pretty awesome. So we get the data safe. And I think we're going to talk about Yamol a little bit later. So

we also going to tell you a little bit about Yamol stuff behind this, but for now, okay, we got data safe. So what's next? I think storage. I think the next thing is networking. Networking. Yeah, I mean, I know that VMware people love the fact that they can assign the network have a VLAN attached to it and then do some, you know, these switches and all of that is really nice and pretty. But you know, as far as I know, a container only has that one IP and a support network and that's really cool. But we, we, we, VMware guys sometimes have some bit a little bit more requirements. How does that match? Yeah, good question. Good question. The thing is, as you just nicely pointed out, I mean, usually if we are just, you know, the usual Kubernetes environment, we are fine with what we have. Usually, or we can always extend it. No, no question. But in a lot of cases, the part network and what you have is already working out quite nicely. And the rest is, you know, usually done with ingress or whatever you need. And then you're fine. But as I said, virtual machine is the thing you usually want to do more with, right? You need more access to it. Exactly. With parts, you know, I usually do microsegmentation with network policies, but you know, sometimes the stateful applications, they are not really there yet. So maybe I just wanted, you know, assigned to a VLAN. So yeah, interesting. Okay. Good question. I mean, that is why I mean, at the end of the day, first of all, kube word has you covered. That is like the easiest answer for that. But at the kube word project itself on GitHub and this occasion, there is like the so called class network add-ons operator, which is then giving your Kubernetes some more networking capabilities. I think the two I want to highlight, which are included in there is multis and NM state and both will help you to, especially multis will help you to add more network, like additional networks to your pod network, existing pod network. And NM state is making sure that your nodes are ready to do something like net like bridges bonding or whatever you usually do in this case. So the combination of these two can get you pretty far already. So that's for example, part of the class and network add-ons operator. They're even more in there. If you're interested in this topic, I highly recommend you to take a look into that. But these two again, I usually very well used in this occasion. And I think you said multis also works for pod. So in case of you have a situation where you want to have a pod with multiple networks, maybe also looking at just for your

Kubernetes workloads. But now we use it for the virtual machines as well. Yeah, correct, correct. And because again, this

Slide 38 (at 1032.0s)



Timestamp: 1032.0s



YouTube Link: <https://www.youtube.com/watch?t=1032>



Slide Image

SEP 09 - 11, 2025
**CONTAINER
days
CONFERENCE**

From VMware to KubeVirt: A
Technical Migration Journey

**ALEXANDER SCHABER
AND RENE SCHRAMOWSKI**
DACHS IT & Red Hat

Networking

DACHS IT | Red Hat



Transcript

data is even more crucial and more baked into that is even more more important. And you can see just a very short one as I just mentioned all the CSI capabilities. You just have to check what is your vendor of choice for storage currently. And the best case is maybe some of the bigger vendors or namely vendors you know and they have well maintained CSI driver, you can just leverage. And in most cases, they are already ready for that. They already offer wheat white money. They already offer snapshotting and what so not. That's pretty awesome. So we get the data safe. And I think we're going to talk about

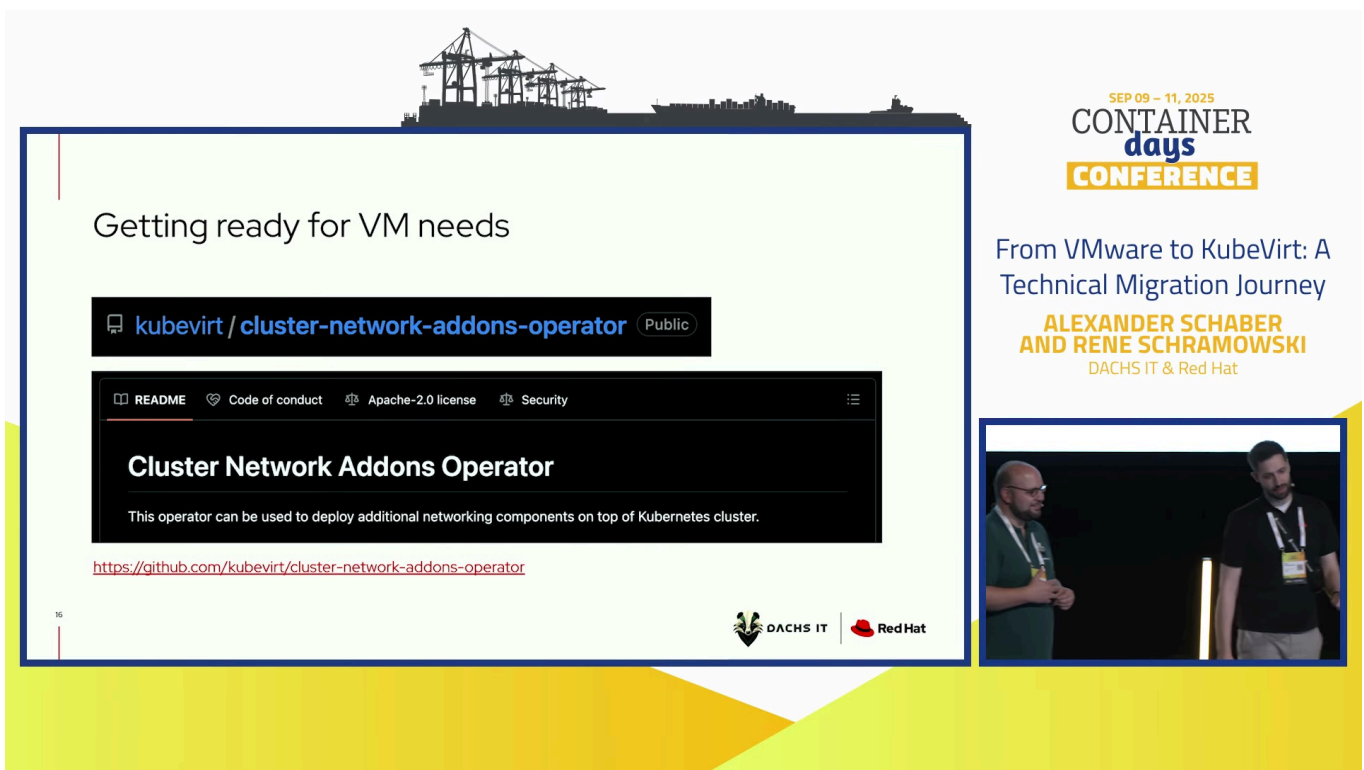
Yamol a little bit later. So we also going to tell you a little bit about Yamol stuff behind this, but for now, okay, we got data safe. So what's next? I think storage. I think the next thing is networking. Networking. Yeah, I mean, I know that VMware people love the fact that they can assign the network have a VLAN attached to it and then do some, you know, these switches and all of that is really nice and pretty. But you know, as far as I know, a container only has that one IP and a

Slide 39 (at 1100.0s)

 **Timestamp:** 1100.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=1100>

 **Slide Image**



Getting ready for VM needs

kubevirt / **cluster-network-addons-operator** Public

README Code of conduct Apache-2.0 license Security

Cluster Network Addons Operator

This operator can be used to deploy additional networking components on top of Kubernetes cluster.

<https://github.com/kubevirt/cluster-network-addons-operator>

DACHS IT Red Hat

SEP 09 - 11, 2025
CONTAINER days
CONFERENCE

From VMware to KubeVirt: A Technical Migration Journey

ALEXANDER SCHABER
AND RENE SCHRAMOWSKI
DACHS IT & Red Hat

 **Transcript**

They already offer wheat white money. They already offer snapshotting and what so not. That's pretty awesome. So we get the data safe. And I think we're going to talk about Yamol a little bit later. So we also going to tell you a little bit about

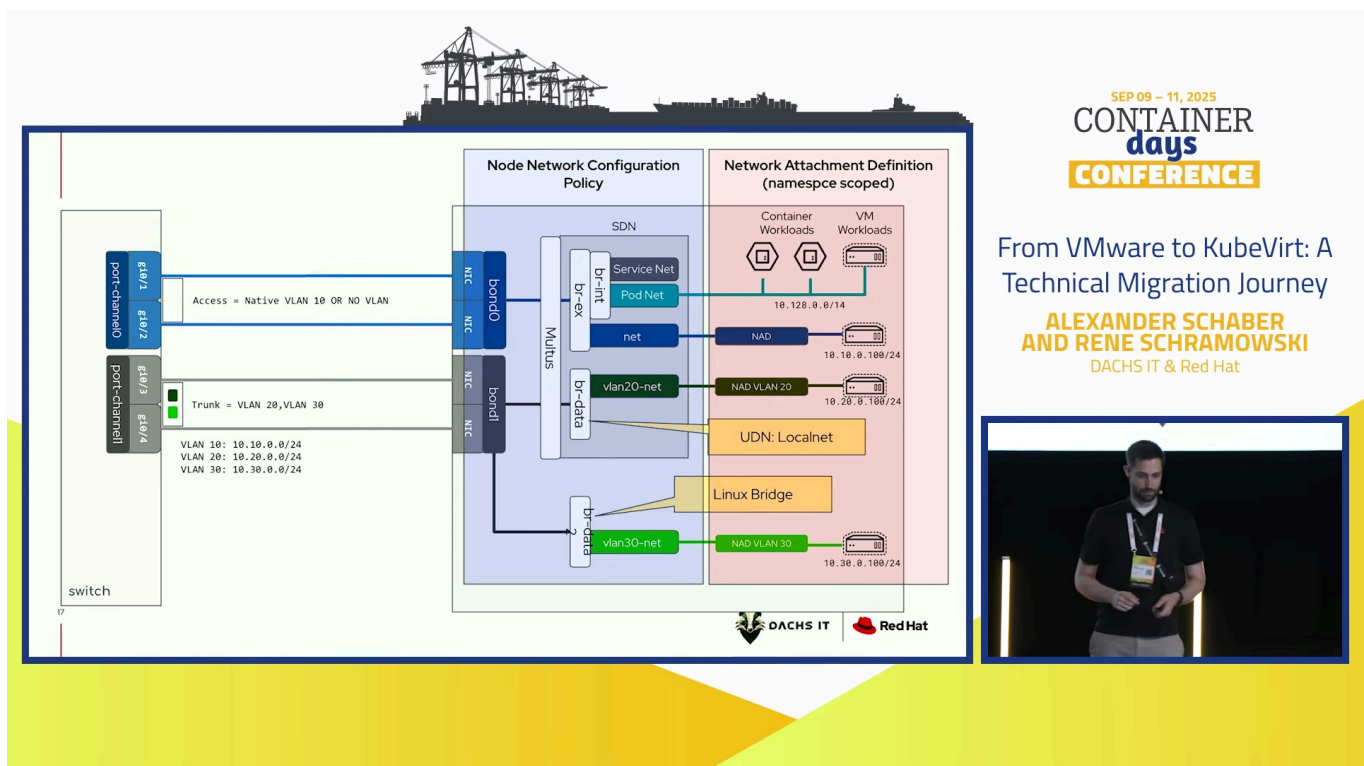
Yamol stuff behind this, but for now, okay, we got data safe. So what's next? I think storage. I think the next thing is networking. Networking. Yeah, I mean, I know that VMware people love the fact that they can assign the network have a VLAN attached to it and then do some, you know, these switches and all of that is really nice and pretty. But you know, as far as I know, a container only has that one IP and a support network and that's really cool. But we, we, we, VMware guys sometimes have some bit a little bit more requirements. How does that match? Yeah, good question. Good question. The thing is, as you just nicely pointed out, I mean, usually if we are just, you know, the usual Kubernetes environment, we are fine with what we have. Usually, or we can always

Slide 40 (at 1162.0s)

 **Timestamp:** 1162.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=1162>

 **Slide Image**



 **Transcript**

the fact that they can assign the network have a VLAN attached to it and then do some, you know, these switches and all of that is really nice and pretty. But you know, as far as I know, a container only has that one IP and a support network and that's really cool. But we, we, we, VMware guys sometimes have some bit a little bit more requirements. How does that match? Yeah, good question. Good question. The thing is, as you just nicely pointed out, I mean, usually if we are just, you know, the usual Kubernetes environment, we are fine with what we have. Usually, or we can always extend it. No, no question. But in a lot of cases, the part network and what you have is already working out quite nicely. And the rest is, you know, usually done with ingress or whatever you need. And then you're fine. But as I said, virtual machine is the thing you usually want to do more with, right? You need more access to it. Exactly. With parts, you know, I usually do micrasegmentation with network policies, but you know, sometimes the stateful applications, they are not really there yet. So maybe I just wanted, you know, assigned to a VLAN. So yeah, interesting. Okay. Good question. I mean, that is why I mean, at the end of the day, first of all, cube word has you covered. That is like the easiest answer for that. But at the cube word project itself on GitHub and this occasion, there is like the so called class network add-ons operator, which is then giving your Kubernetes some more networking capabilities. I think the two I want to highlight, which are included in there is multis

Slide 41 (at 1256.0s)



Timestamp: 1256.0s



YouTube Link: <https://www.youtube.com/watch?t=1256>



Slide Image



Transcript

just nicely pointed out, I mean, usually if we are just, you know, the usual Kubernetes environment, we are fine with what we have. Usually, or we can always extend it. No, no question. But in a lot of cases, the part network and what you have is already working out quite nicely. And the rest is, you know, usually done with ingress or whatever you need. And then you're fine. But as I said, virtual machine is the thing you usually want to do more with, right? You need more access to it. Exactly. With parts, you know, I usually do micrasegmentation with network policies, but you know, sometimes the stateful applications, they are not really there yet. So maybe I

Slide 42 (at 1295.0s)



Timestamp: 1295.0s



YouTube Link: <https://www.youtube.com/watch?t=1295>



Slide Image



SEP 09 – 11, 2025
CONTAINER
days
CONFERENCE

From VMware to KubeVirt: A
Technical Migration Journey

ALEXANDER SCHABER
AND RENE SCHRAMOWSKI
DACHS IT & Red Hat

```
0 # k get VirtualMachineClusterPreference.rhel.10 -o yaml
1 apiVersion:instancetype.kubevirt.io/v1beta1
2 kind:VirtualMachineClusterPreference
3 metadata:
4   name: rhel.10
5 spec:
6   annotations:
7     vm.kubevirt.io/os: linux
8   devices:
9     preferredDiskBus: virtio
10    preferredDiskDedicatedIoThread: true
11    preferredInterfaceModel: virtio
12    preferredRng: {}
13  features:
14    preferredSmm: {}
15  firmware:
16    preferredEfi:
17      secureBoot: true
18  requirements:
19    cpu:
20      guest: 1
21    memory:
22      guest: 1536Mi
```

19

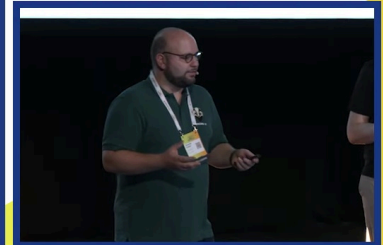
<https://github.com/kubevirt/common-instancetype>



DACHS IT



Red Hat



Transcript

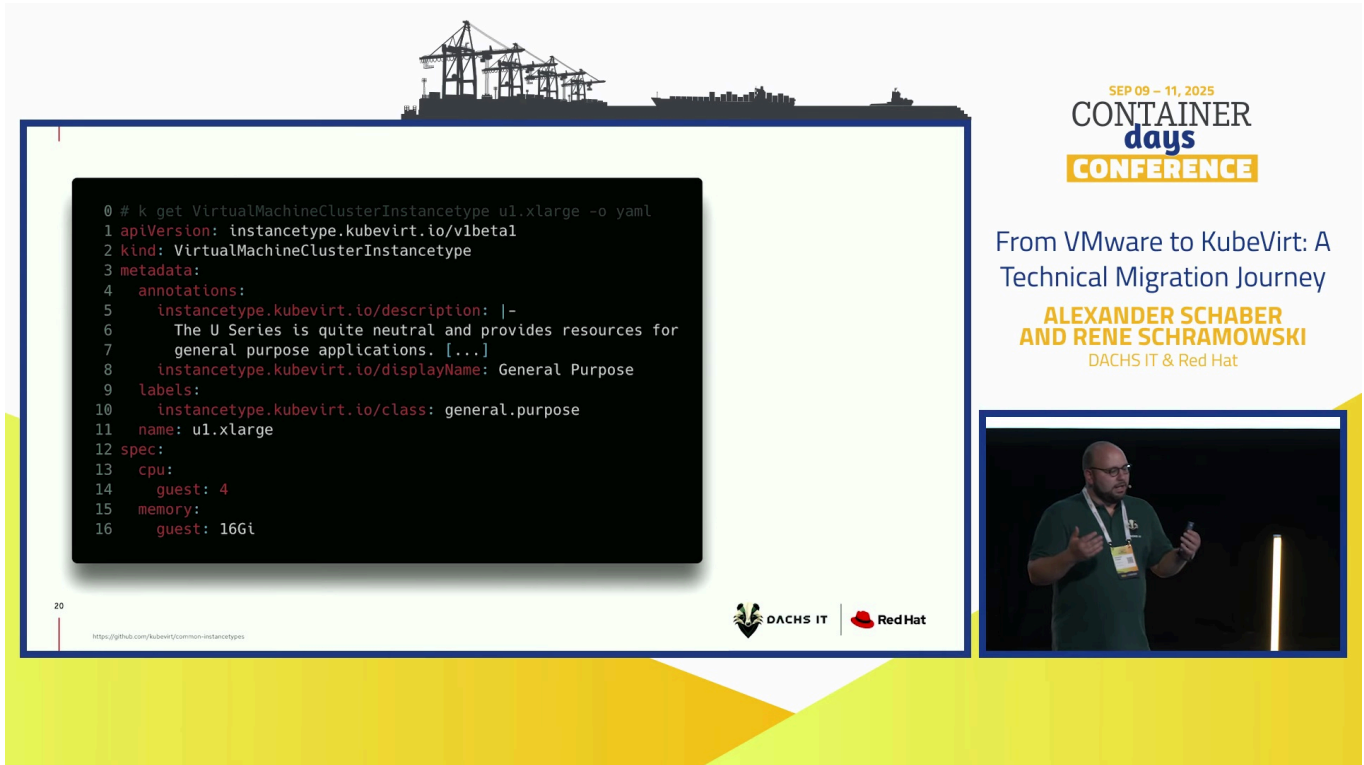
want to do more with, right? You need more access to it. Exactly. With parts, you know, I usually do micrasegmentation with network policies, but you know, sometimes the stateful applications, they are not really there yet. So maybe I just wanted, you know, assigned to a VLAN. So yeah, interesting. Okay. Good question. I mean, that is why I mean, at the end of the day, first of all, kube word has you covered. That is like the easiest answer for that. But at the kube word project itself on GitHub and this occasion, there is like the so called class network add-ons operator, which is then giving your Kubernetes some more networking capabilities. I think the two I want to highlight, which are included in there is multis and NM state and both will help you to, especially multis will help you to add more network, like additional networks to your pot network, existing pot network. And NM state is making sure that your notes are ready to do something like net like bridges bonding or whatever you usually do in this case. So the combination of these two can get you pretty far already. So that's for example, part of the class and network add-ons operator. They're even more in there. If you're interested in this topic, I highly recommend you to take a look into

Slide 43 (at 1368.0s)

 **Timestamp:** 1368.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=1368>

Slide Image



The slide features a dark-themed terminal window on the left displaying a Kubernetes YAML configuration for a VirtualMachineClusterInstanceType. The configuration includes metadata, annotations, and labels. On the right, there is a promotional banner for the Container Days Conference, dated September 9-11, 2025, with the title 'From VMware to KubeVirt: A Technical Migration Journey' by Alexander Schaber and Rene Schramowski. A small inset photo shows a speaker at a podium. Logos for DACHS IT and Red Hat are visible at the bottom.

```
0 # k get VirtualMachineClusterInstanceType u1.xlarge -o yaml
1 apiVersion:instancetype.kubevirt.io/v1beta1
2 kind: VirtualMachineClusterInstanceType
3 metadata:
4   annotations:
5     instancetype.kubevirt.io/description: |-
6       The U Series is quite neutral and provides resources for
7       general purpose applications. [...]
8     instancetype.kubevirt.io/displayName: General Purpose
9   labels:
10    instancetype.kubevirt.io/class: general.purpose
11   name: u1.xlarge
12 spec:
13   cpu:
14     guest: 4
15   memory:
16     guest: 16Gi
```

20
<https://github.com/kubevirt/common-instancetype>

SEP 09 – 11, 2025
CONTAINER days CONFERENCE

From VMware to KubeVirt: A Technical Migration Journey

ALEXANDER SCHABER AND RENE SCHRAMOWSKI
DACHS IT & Red Hat

DACHS IT | Red Hat

Transcript

But at the cube word project itself on GitHub and this occasion, there is like the so called class network add-ons operator, which is then giving your Kubernetes some more networking capabilities. I think the two I want to highlight, which are included in there is multis and NM state and both will help you to, especially multis will help you to add more network, like additional networks to your pot network, existing pot network. And NM state is making sure that your notes are ready to do something like net

Slide 44 (at 1397.0s)

 **Timestamp:** 1397.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=1397>



Slide Image



Transcript

your notes are

Slide 45 (at 1398.0s)



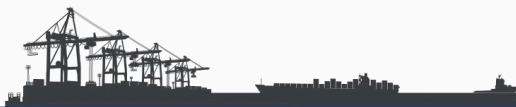
Timestamp: 1398.0s



YouTube Link: <https://www.youtube.com/watch?t=1398>



Slide Image



SEP 09 – 11, 2025
**CONTAINER
days
CONFERENCE**

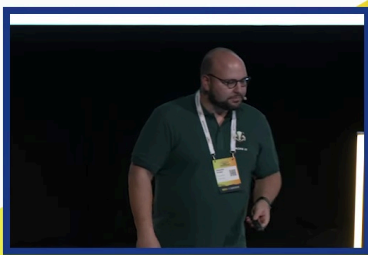
From VMware to KubeVirt: A
Technical Migration Journey

**ALEXANDER SCHABER
AND RENE SCHRAMOWSKI**
DACHS IT & Red Hat

```
0 # virtctl create vm --instancetype=u1.xlarge --preference=rhel.10
1 apiVersion: kubevirt.io/v1
2 kind: VirtualMachine
3 metadata:
4   name: vm-zvngw
5 spec:
6   instancetype:
7     name: u1.xlarge
8   preference:
9     name: rhel.10
10  template:
11    spec:
12      domain:
13        devices:
14          interfaces:
15            - name: default
16              masquerade: {}
17          resources: {}
18        networks:
19          - name: default
20            pod: {}
```

21

DACHS IT | Red Hat



Transcript

also works for pod. So in case of you have a situation where you want to have a pod with multiple networks, maybe also looking at just for your Kubernetes workloads. But now we use it for the virtual machines as well. Yeah, correct, correct. And because again, this picture is like a little bit crowded and I don't want to focus too much into it. But there you can see some of the things I mentioned. There's this multis, for example, to enable their additional networking. And usually when you're coming from a visualization world, you want to be able to kind of expose like a fully virtual machine to different protocols on specific switches and what are what not. And you usually make use of the VLAN tagging scenarios and what not. And short hand story because time is crucial here. This thing got you covered again, they only the combination of motors and NM state will get you really far already. And I've seen implementation. Like for example, with the moment with the NM states, you will be you're going to get some custom resources to do the additional network configuration or the attachments. And then you are able to kind of emulate what you're used to from your virtualization world to kind of be able to, for example, do VLAN tagging a complete machine on a specific port to your physical network. And then kind of emulate what you have been used before. This is kind

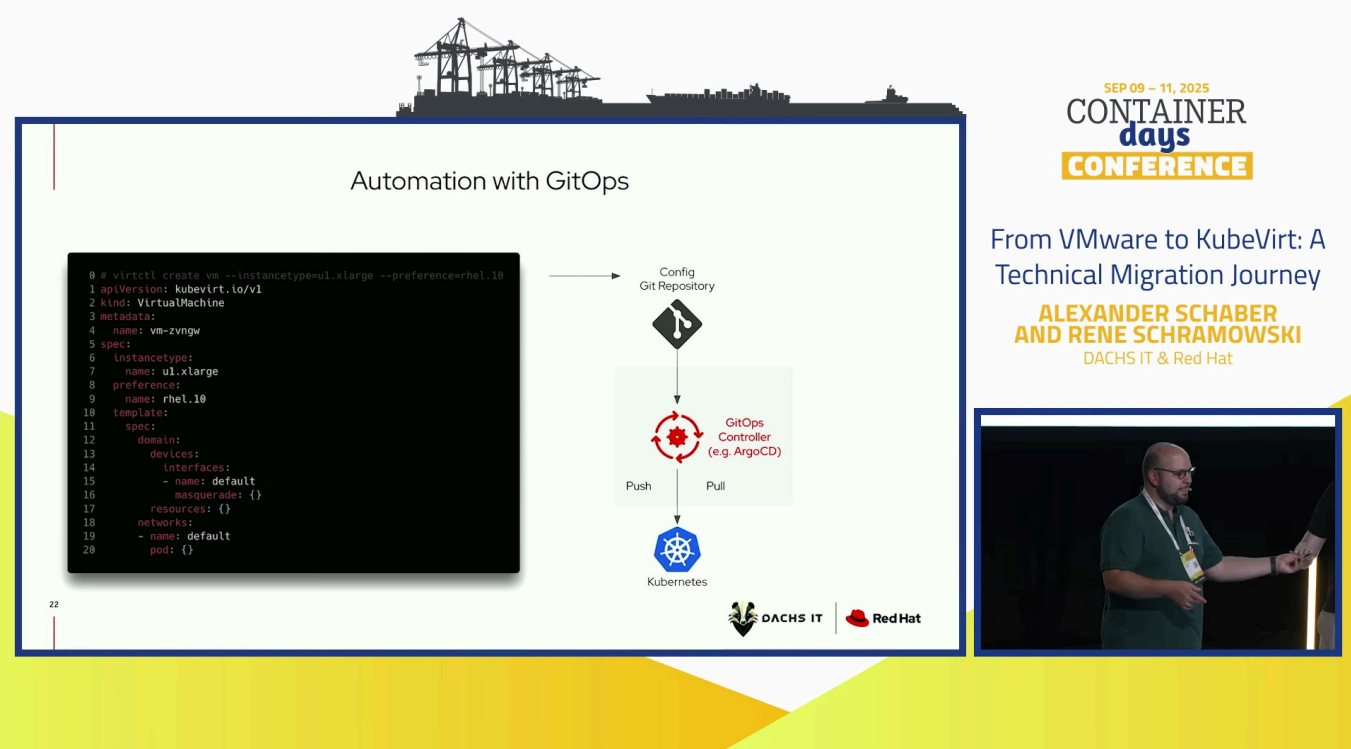
of short answer just because of the time, but we can of course talk about later on. But at the end of the result is there is something for it. And you can kind of do it on Kubernetes as well. That's cool. All right. I mean, if you might. All right. Let's get the Yamlink. I mean, one of the biggest differences between, you know, maybe your current setup

Slide 46 (at 1500.0s)

 **Timestamp:** 1500.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=1500>

Slide Image



Automation with GitOps

```
0 # virtctl create vm --instancetype=u1.xlarge --preference=rhel.10
1 apiVersion: kubevirt.io/v1
2 kind: VirtualMachine
3 metadata:
4   name: vm-zvngw
5 spec:
6   instancetype:
7     name: u1.xlarge
8   preference:
9     name: rhel.10
10  template:
11    spec:
12      domain:
13        devices:
14          interfaces:
15            - name: default
16              masquerade: {}
17          resources: {}
18      networks:
19        - name: default
20          pod: {}
```

Config Git Repository

Push

GitOps Controller (e.g. ArgoCD)

Pull

Kubernetes

SEP 09 - 11, 2025
CONTAINER days
CONFERENCE

From VMware to KubeVirt: A Technical Migration Journey

ALEXANDER SCHABER
AND RENE SCHRAMOWSKI
DACHS IT & Red Hat

Transcript

enable their additional networking. And usually when you're coming from a visualization world, you want to be able to kind of expose like a fully virtual machine to different protocols on specific switches and what are what not. And you usually make use of the VLAN tagging scenarios and what not. And short hand story because time is crucial here. This thing got you covered again, they

only the combination of motors and NM state will get you really far already. And I've seen implementation. Like for example, with the moment with the NM states, you will be you're going to get some custom resources to do the additional network configuration or the attachments. And then you are able to kind of emulate what you're used to from your virtualization world to kind of be able to, for example, do VLAN tagging a complete machine on a specific port to your physical network. And then kind of emulate what you have been used before. This is kind of short answer just because of the time, but we can of course talk about later on. But at the end of the result is there is something for it. And you can kind of do it on Kubernetes as well. That's cool. All right. I mean, if you might. All right. Let's get the Yamlink. I mean, one of the biggest differences between, you know, maybe your current setup where you have VMware a lot of

Slide 47 (at 1578.0s)

 **Timestamp:** 1578.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=1578>

 **Slide Image**



SEP 09 – 11, 2025
**CONTAINER
days
CONFERENCE**

From VMware to KubeVirt: A
Technical Migration Journey

**ALEXANDER SCHABER
AND RENE SCHRAMOWSKI**
DACHS IT & Red Hat

23

DACHS IT | Red Hat



Transcript

I've seen implementation. Like for example, with the moment with the NM states, you will be you're going to get some custom resources to do the additional network configuration or the attachments. And then you are able to kind of emulate what you're used to from your virtualization world to kind of be able to, for example, do VLAN tagging a complete machine on a specific port to your physical network. And then kind of emulate what you have been used before. This is kind of short answer just because of the time, but we can of course talk about later on. But at the end of the result is there is something for it. And you can kind of do it on Kubernetes as well. That's cool. All right. I mean, if you might. All right. Let's get the Yamlink. I mean, one of the biggest differences between, you know, maybe your current setup where you have VMware a lot of click ups. Maybe you build some automation around it. I mean, there are applications like we realize or even some open source applications you can use to automate your VMware processes. It's really interesting to see how that gets translated into the new Yamal world of, you know, everything is a declarative state. So maybe let's have a look at a few custom resources that Cupid brings to the table and gives you the opportunity to standardize maybe even your virtualization setup. The first one I want to show you. I think it's legible. I hope I hope it's also legible from the last row is there is a custom resource type and they're all a mouthful. It's called virtual machine cluster preference. And here you can see that I can set specific settings for a certain operating system. So for instance, in our example, we took a rail 10 and rail 10 supports secure boot three feet. So here we can say please for all virtual machines which are based on rail 10. If I want to use this preference, I get automatically enabled secure boot three feet. And there's a bunch of them already pre-rendered for you. So, you know, also for windows and other other operating systems. And that gets you going really good really well because you don't have to consider what do I need for each operating system. And also it gets you standardization if you have this as a preference, all of them will be the same. So, and you can see there are some requirements. You know, a virtual machine with rail should have at least one CPU and 1.5 gigabytes of memory. But obviously that's not every workload. So, here's another thing that comes into play is called virtual machine cluster

instance type, I guess that rolls right off. And here I wanted to show you an example. It's called U1 X large. And it just specifies hey, this is a general purpose of virtual machine. And it comes with a standard or very typical one to four ratio. So you get four CPUs and six and gigabytes of memory. So, we got the operating system, we got our sizing. Let's bring them together. And there's actually a tool that helps you create virtual machine manifest. It's called virtual. And I said, hey, listen, virtual. I want to create a VM. It should have you want X large as sizing. And it should be a well 10 preference. And what it gives me is basically a YAML manifest, which I then, you know, can work on and extend, which I actually did. So, I'm going to get to this in a moment. And you can see online six to nine that my preferences and my operating system are now already referenced, which means that I don't no longer need to care about sizing or those special options. They are basically generalized into that separate custom resource. And it's kind of cool. It makes it way shorter. And that also maybe gets you in a situation where not every virtual machine is different. It gets you to a situation where you have more standardized, more maybe a cloud like situation where you have standardized instance types, which, by the way, also makes it maybe easier to do internal billing because you can, you know, it's not like per megabyte RAM. It's per that instance type is going to cost you that much. It makes it way easier for you. The thing I actually added here, that's not something that virtual out puts automatically is I wanted to add the networking device. So you see from line 13 to 16. Hey, listen, I want a networking device. It should be in the default network. And the default network is online 18 to 20 and just a regular pot networking, but that's where the maltose comes in. That's where you can expand on. And so if I have all that, I got all the gamble money first. Can I do something cool with them? As always, I think a lot of people agree that the awesome thing you can do with your notes is get ups, right? I mean, just as a heads up, like who knows get ups or just a hands up for

Slide 48 (at 1859.0s)

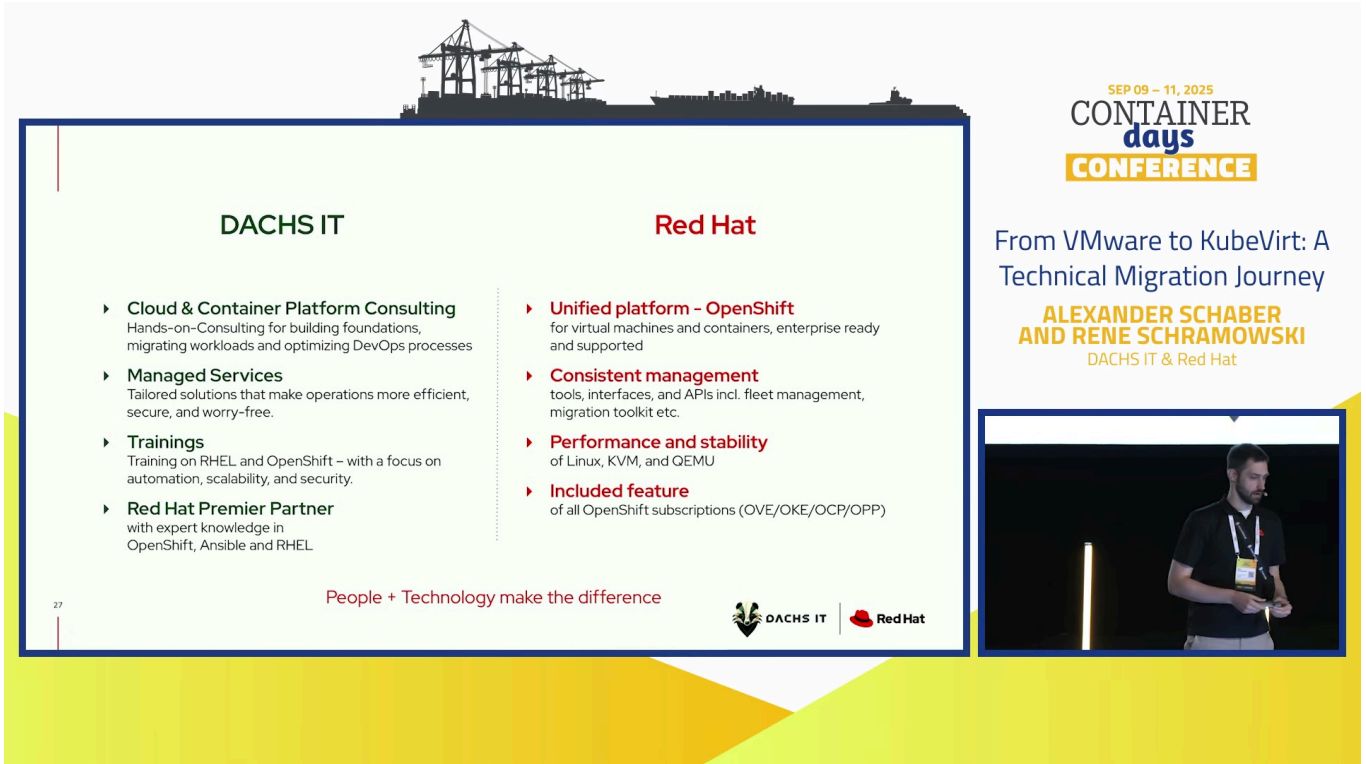


Timestamp: 1859.0s



YouTube Link: <https://www.youtube.com/watch?t=1859>

Slide Image



SEP 09 - 11, 2025
CONTAINER
days
CONFERENCE

From VMware to KubeVirt: A Technical Migration Journey
ALEXANDER SCHABER
AND RENE SCHRAMOWSKI
DACHS IT & Red Hat

DACHS IT

- ▶ **Cloud & Container Platform Consulting**
Hands-on-Consulting for building foundations, migrating workloads and optimizing DevOps processes
- ▶ **Managed Services**
Tailored solutions that make operations more efficient, secure, and worry-free.
- ▶ **Trainings**
Training on RHEL and OpenShift – with a focus on automation, scalability, and security.
- ▶ **Red Hat Premier Partner**
with expert knowledge in OpenShift, Ansible and RHEL.

Red Hat

- ▶ **Unified platform - OpenShift**
for virtual machines and containers, enterprise ready and supported
- ▶ **Consistent management**
tools, interfaces, and APIs incl. fleet management, migration toolkit etc.
- ▶ **Performance and stability**
of Linux, KVM, and QEMU
- ▶ **Included feature**
of all OpenShift subscriptions (OVE/OKE/OCF/OPP)

27

People + Technology make the difference

DACHS IT | **Red Hat**

Transcript

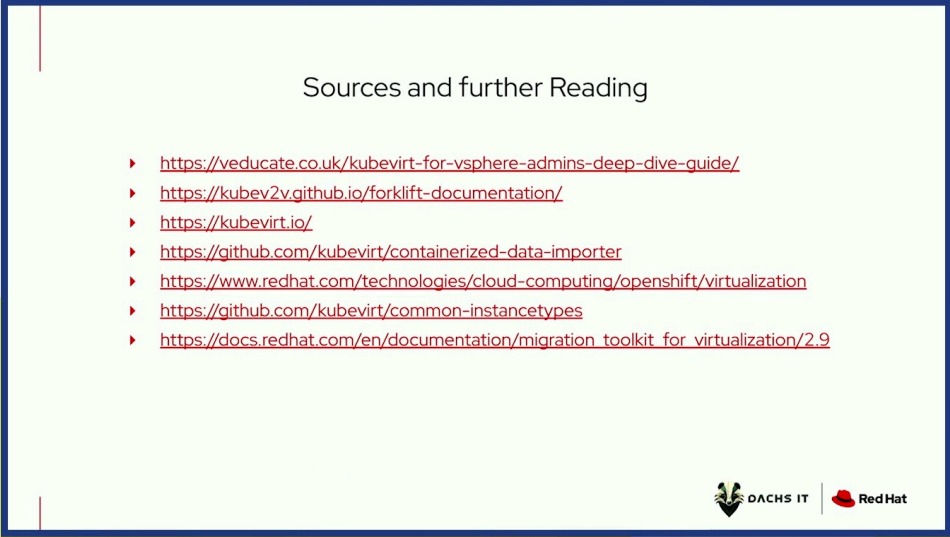
is kind of short answer just because of the time, but we can of course talk about later on. But at the end of the result is there is something for it. And you can kind of do it on Kubernetes as well. That's cool. All right. I mean, if you might. All right. Let's get the Yamlink. I

Slide 49 (at 1878.0s)

 **Timestamp:** 1878.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=1878>

Slide Image



Sources and further Reading

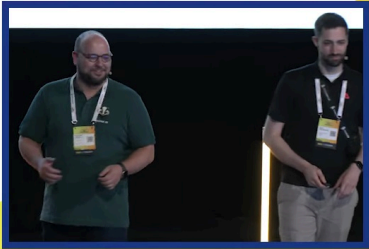
- ▶ <https://veducate.co.uk/kubevirt-for-vsphere-admins-deep-dive-guide/>
- ▶ <https://kubev2v.github.io/forklift-documentation/>
- ▶ <https://kubevirt.io/>
- ▶ <https://github.com/kubevirt/containerized-data-importer>
- ▶ <https://www.redhat.com/technologies/cloud-computing/openshift/virtualization>
- ▶ <https://github.com/kubevirt/common-instancetypes>
- ▶ https://docs.redhat.com/en/documentation/migration_toolkit_for_virtualization/2.9

DACHS IT | Red Hat

SEP 09 – 11, 2025
CONTAINER
days
CONFERENCE

From VMware to KubeVirt: A Technical Migration Journey

ALEXANDER SCHABER
AND RENE SCHRAMOWSKI
DACHS IT & Red Hat



Transcript

you build some automation around it. I mean, there are applications like we realize or even some open

Slide 50 (at 1884.0s)



Timestamp: 1884.0s



YouTube Link: <https://www.youtube.com/watch?t=1884>



Slide Image

SEP 09 – 11, 2025
**CONTAINER
days
CONFERENCE**

From VMware to KubeVirt: A Technical Migration Journey
**ALEXANDER SCHABER
AND RENE SCHRAMOWSKI**
DACHS IT & Red Hat

Thank you

Red Hat is the world's leading provider of enterprise open source software solutions. Award-winning support, training, and consulting services make Red Hat a trusted adviser to the Fortune 500.

in linkedin.com/company/red-hat
youtube.com/user/RedHatVideos
facebook.com/redhatinc
twitter.com/RedHat

29

DACHS IT | Red Hat



Transcript

want to show you. I think it's legible. I hope I hope it's also legible from the last

Slide 51 (at 1890.0s)



Timestamp: 1890.0s



YouTube Link: <https://www.youtube.com/watch?t=1890>



Slide Image



Transcript

which are based on rail 10. If I want to use this preference, I get automatically enabled secure boot three feet. And there's a bunch of them already pre-rendered for you. So, you know,

Slide 52 (at 1901.0s)



Timestamp: 1901.0s



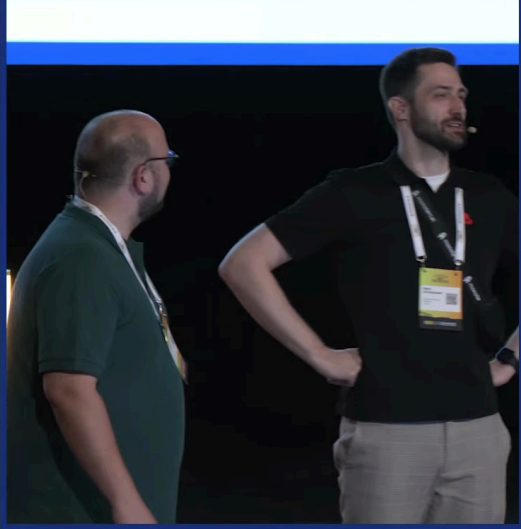
YouTube Link: <https://www.youtube.com/watch?t=1901>



Slide Image



SEP 09 – 11, 2025
CONTAINER
days
CONFERENCE



Lisa9

As you already mentioned in your presentation, normally it's 2 different teams that handle VMs and k8s, how would this work in the real world? Merge them? Teach?

Anonymous6

What's the name of your mascot? 🐾

DACHS IT6

Nice try! But the badger's name will only be revealed after tomorrow's raffle at 1:30pm at our booth! 🐾

Leon6

How hard is it to recreate KubeVirt VMs, just do a „kubectl rollout restart“? Are they rather pets or cattle?

Latest question

Anonymous0

Are there any optimizations available to mitigate large container images when migrating from VMs?

Transcript

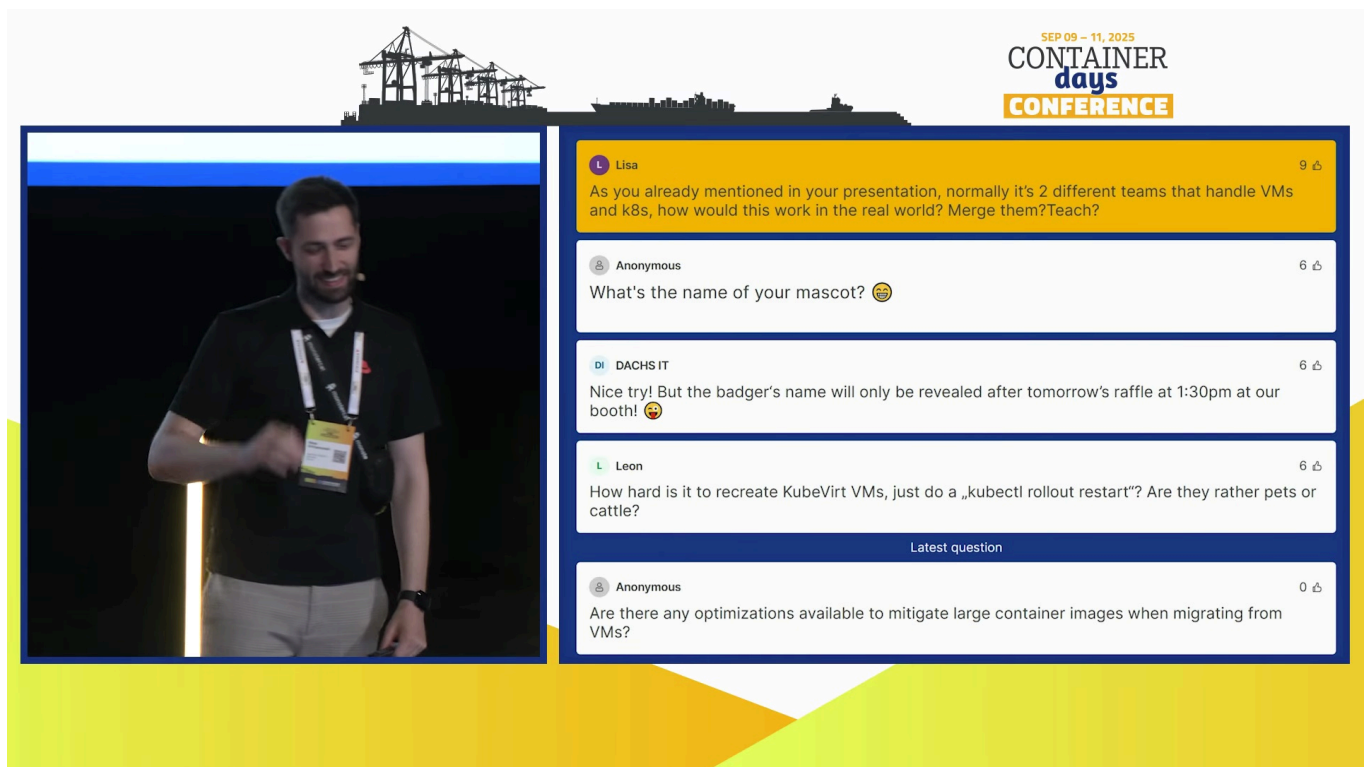
the same. So,

Slide 53 (at 1902.0s)

 **Timestamp:** 1902.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=1902>

 **Slide Image**



Transcript

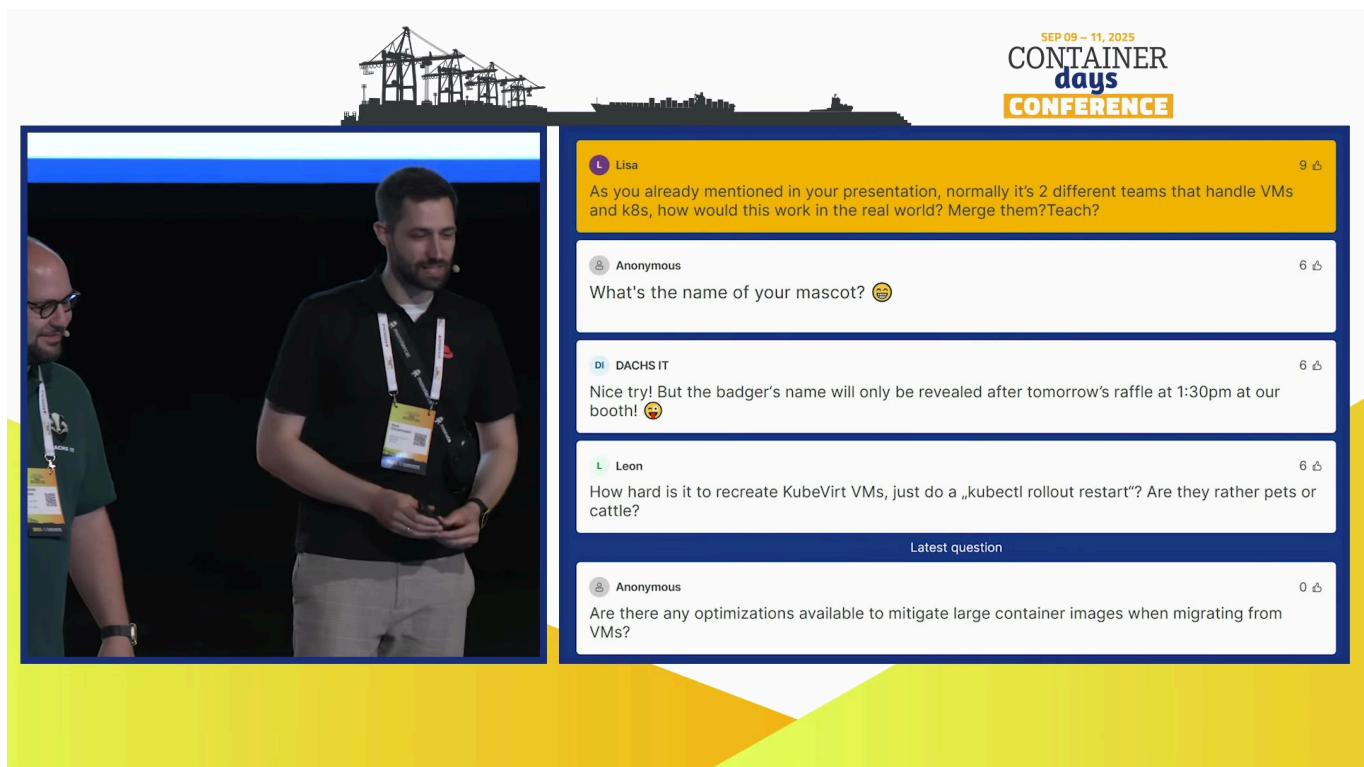
machine. And it comes with a standard or very typical one to four ratio. So

Slide 54 (at 1907.0s)

 **Timestamp:** 1907.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=1907>

Slide Image



Transcript

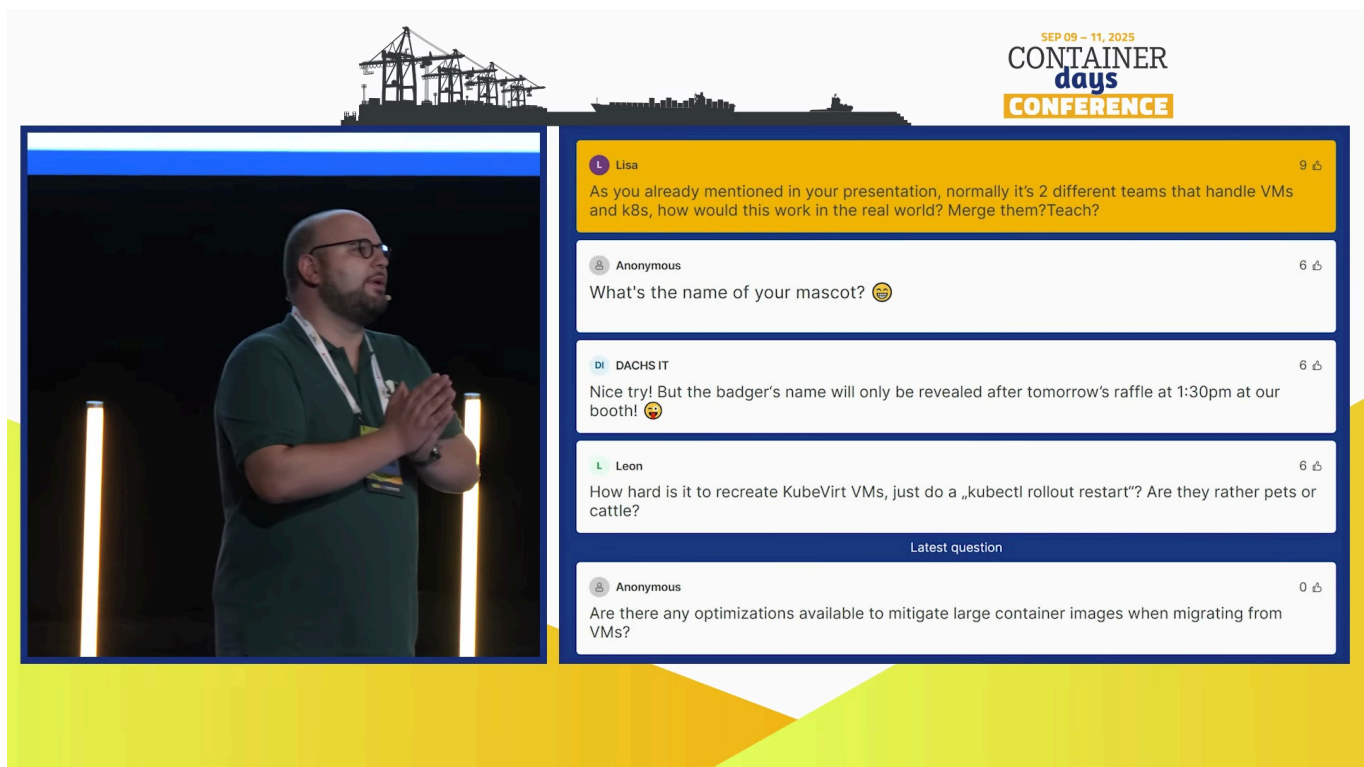
And what it gives me is basically a YAML manifest, which I then, you know, can work on and extend, which I actually did. So, I'm going to get to this in a moment. And you can see online six to nine that my preferences and my operating system are now already referenced, which means that I don't no longer need to care about sizing or those special options. They are basically generalized into that separate custom resource.

Slide 55 (at 1932.0s)

 **Timestamp:** 1932.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=1932>

 **Slide Image**



Transcript

It makes it way shorter. And that also maybe gets you in a situation where not every virtual machine is different. It gets you to a situation where you have more standardized, more maybe a cloud like situation where you have standardized instance types, which, by the way, also makes it maybe easier to do internal billing because you can, you know, it's not like per megabyte RAM. It's per that instance type is going to cost you that much. It makes it way easier for you. The thing I actually added here, that's not something that virtual out puts automatically is I wanted to add the networking device. So you see from line 13 to 16. Hey, listen, I want a networking device. It should be in the default network. And the default

Slide 56 (at 1975.0s)



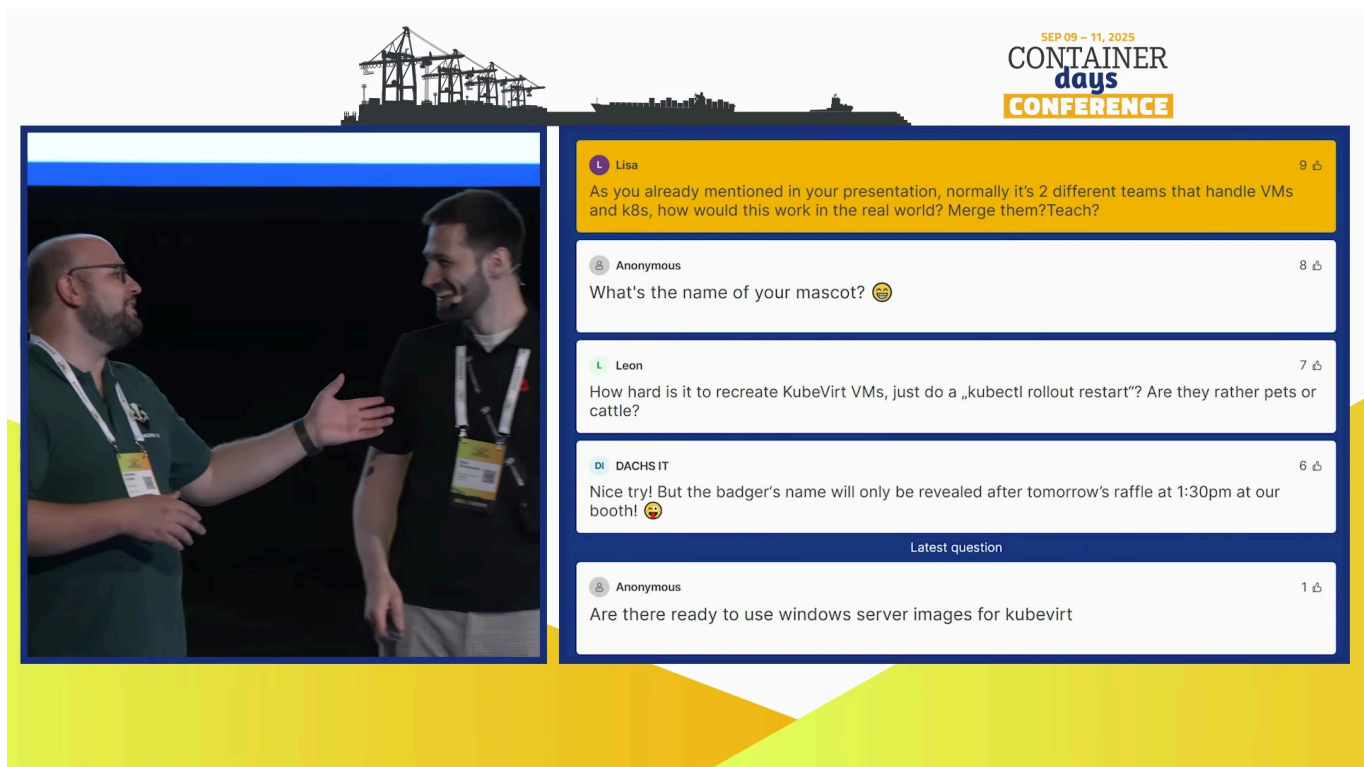
Timestamp: 1975.0s



YouTube Link: <https://www.youtube.com/watch?t=1975>



Slide Image



Transcript

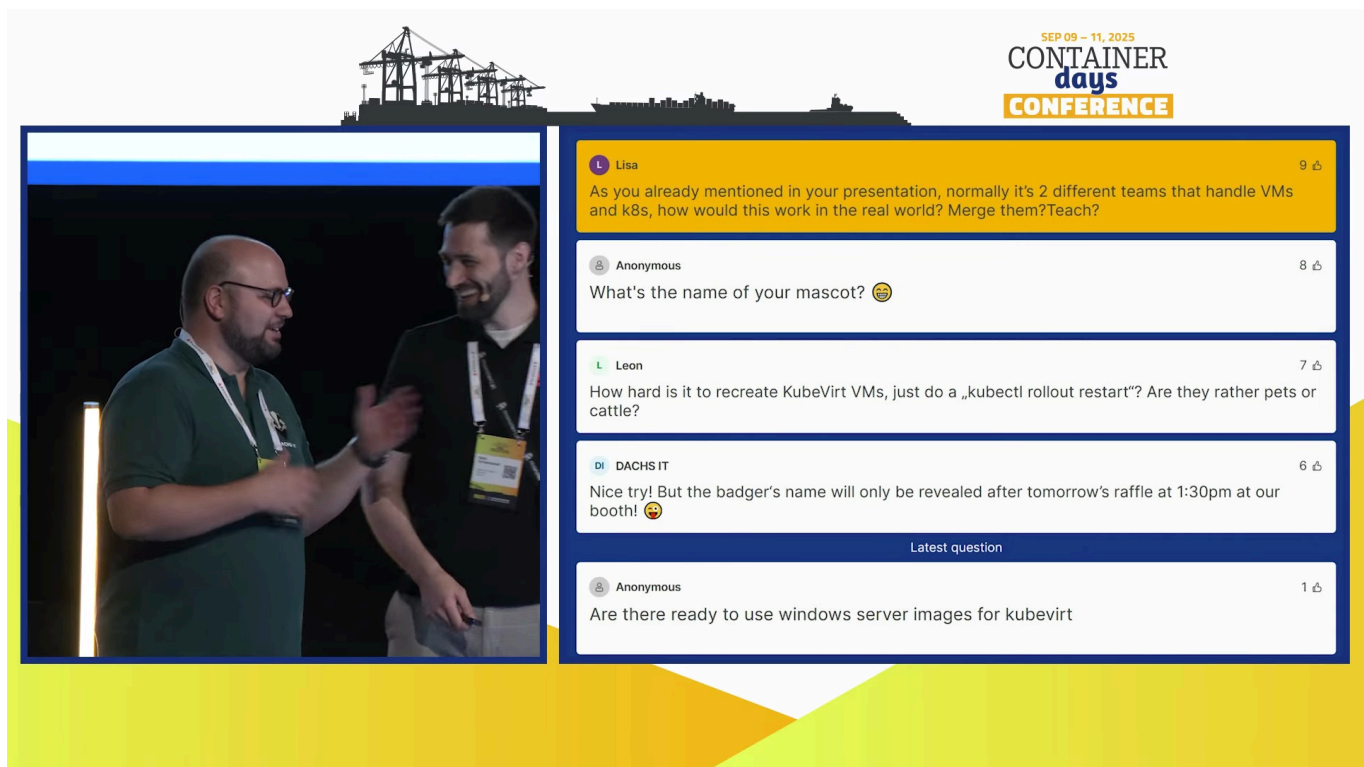
it way easier

Slide 57 (at 1976.0s)

 **Timestamp:** 1976.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=1976>

 **Slide Image**



Transcript

that, I got

Slide 58 (at 1977.0s)



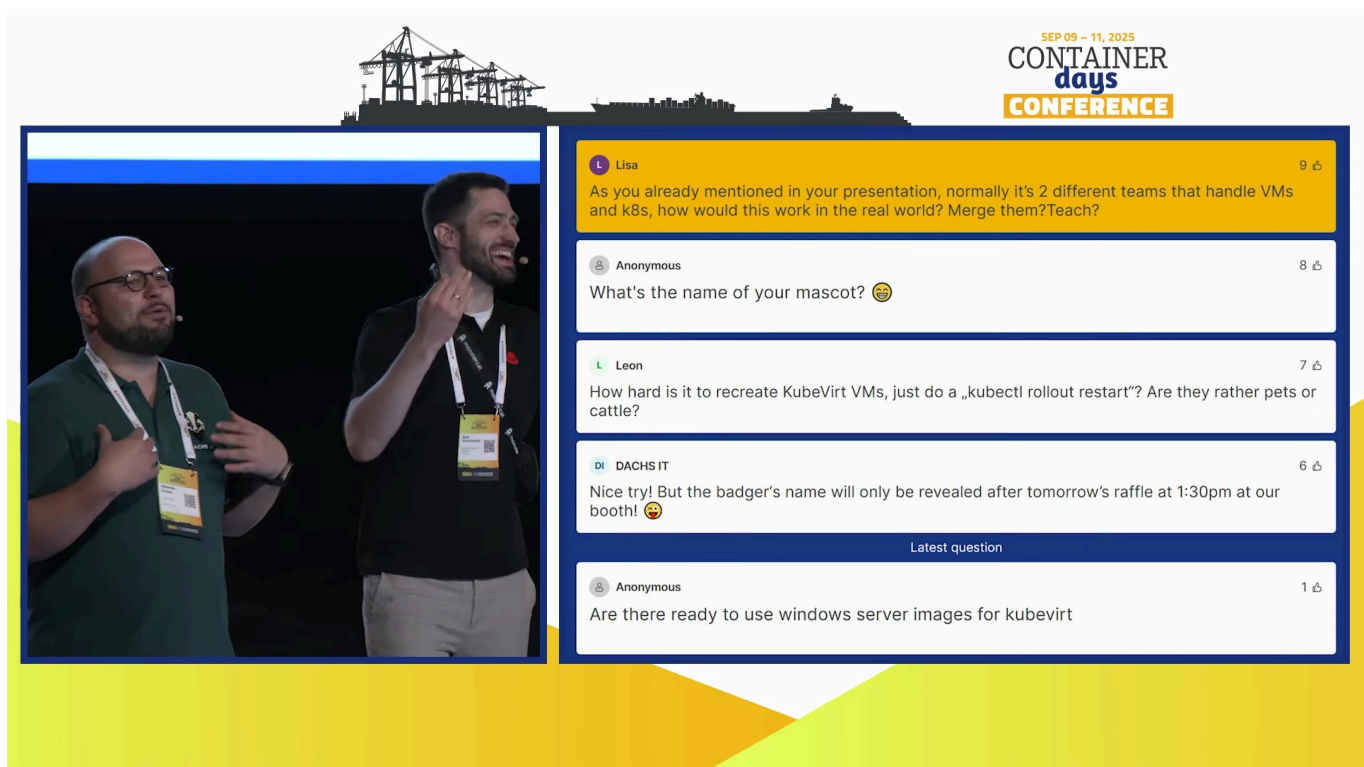
Timestamp: 1977.0s



YouTube Link: <https://www.youtube.com/watch?t=1977>



Slide Image



Transcript

we can know

Slide 59 (at 1978.0s)



Timestamp: 1978.0s



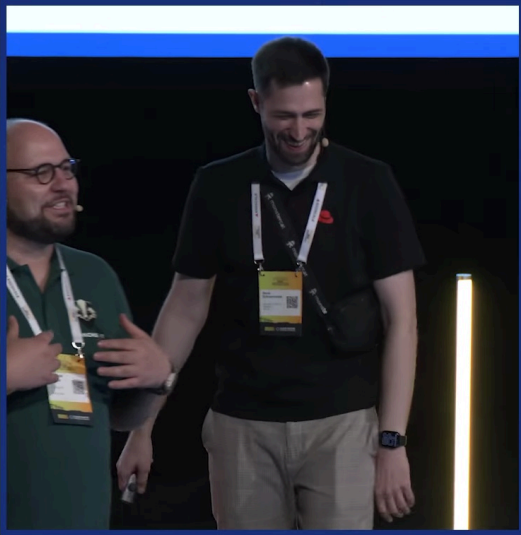
YouTube Link: <https://www.youtube.com/watch?t=1978>



Slide Image



SEP 09 – 11, 2025
CONTAINER
days
CONFERENCE



Lisa9

As you already mentioned in your presentation, normally it's 2 different teams that handle VMs and k8s, how would this work in the real world? Merge them? Teach?

Anonymous8

What's the name of your mascot? 🐾

Leon7

How hard is it to recreate KubeVirt VMs, just do a „kubect! rollout restart“? Are they rather pets or cattle?

DACHS IT6

Nice try! But the badger's name will only be revealed after tomorrow's raffle at 1:30pm at our booth! 🐾

Latest question

Anonymous1

Are there ready to use windows server images for kubevirt



Transcript

of granted here

Slide 60 (at 1979.0s)



Timestamp: 1979.0s



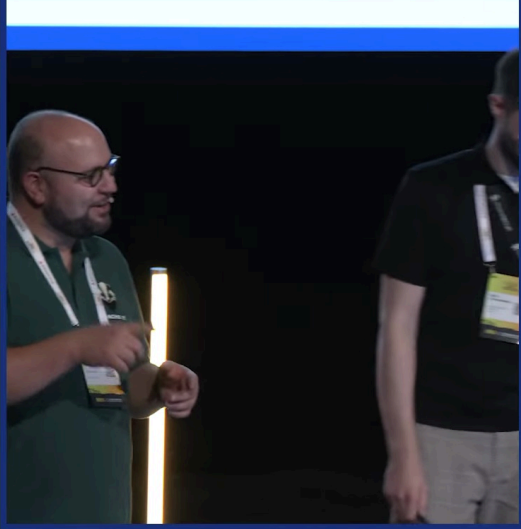
YouTube Link: <https://www.youtube.com/watch?t=1979>



Slide Image



SEP 09 – 11, 2025
CONTAINER
days
CONFERENCE



Anonymous 8

What's the name of your mascot? 🐾

Leon 7

How hard is it to recreate KubeVirt VMs, just do a „kubectl rollout restart“? Are they rather pets or cattle?

DACHS IT 6

Nice try! But the badger's name will only be revealed after tomorrow's raffle at 1:30pm at our booth! 🐼

Sebastian 5

Do you experience problems running containers for days/weeks/months without restarting (because you don't want to restart VMs every few hours)?

Latest question

Anonymous 1

Are there ready to use windows server images for kubevirt



Transcript

were very impative and somebody needs to click and do something with get ups, you know, you might have to rethink a little bit those processes. But we believe that actually it's a change for the better because you get more into a declarative state and go away from an impative configuration. 100% OK, so now for the last part, a little bit of the migration itself, at least to give you some idea or some hints of what to do and how to tackle this topic. Well, we kind of said it's a transformation journey, right? And we wanted to give you all the basics. So when you start your journey, you have, you know, the terminology down, you know what they were talking about, those different tools. So I hope we got that. And now let's talk a little bit about what to actually do. So it's easy, right? It's just right the animals, right? So we have just seen it.

Slide 61 (at 2031.0s)



Timestamp: 2031.0s



YouTube Link: <https://www.youtube.com/watch?t=2031>



Slide Image

SEP 09 - 11, 2025
CONTAINER
days
CONFERENCE

Q&A Section:

- Sebastian** (5 upvotes): Do you experience problems running containers for days/weeks/months without restarting (because you don't want to restart VMs every few hours)?
- Anonymous** (4 upvotes): What security capabilities are needed to run KVM via kubevirt in a container? Are full privileged containers needed?
- Anonymous** (3 upvotes): How do you bootstrap a KubeVirt cluster when you want to run k8s on those VMs managed by KubeVirt?
- Anonymous** (3 upvotes): Is there a UI to ease the transition for the Sysadmins used to vSphere that are less comfortable with YAML?
- Latest question**
- Anonymous** (2 upvotes): Are there ready to use windows server images for kubevirt



Transcript

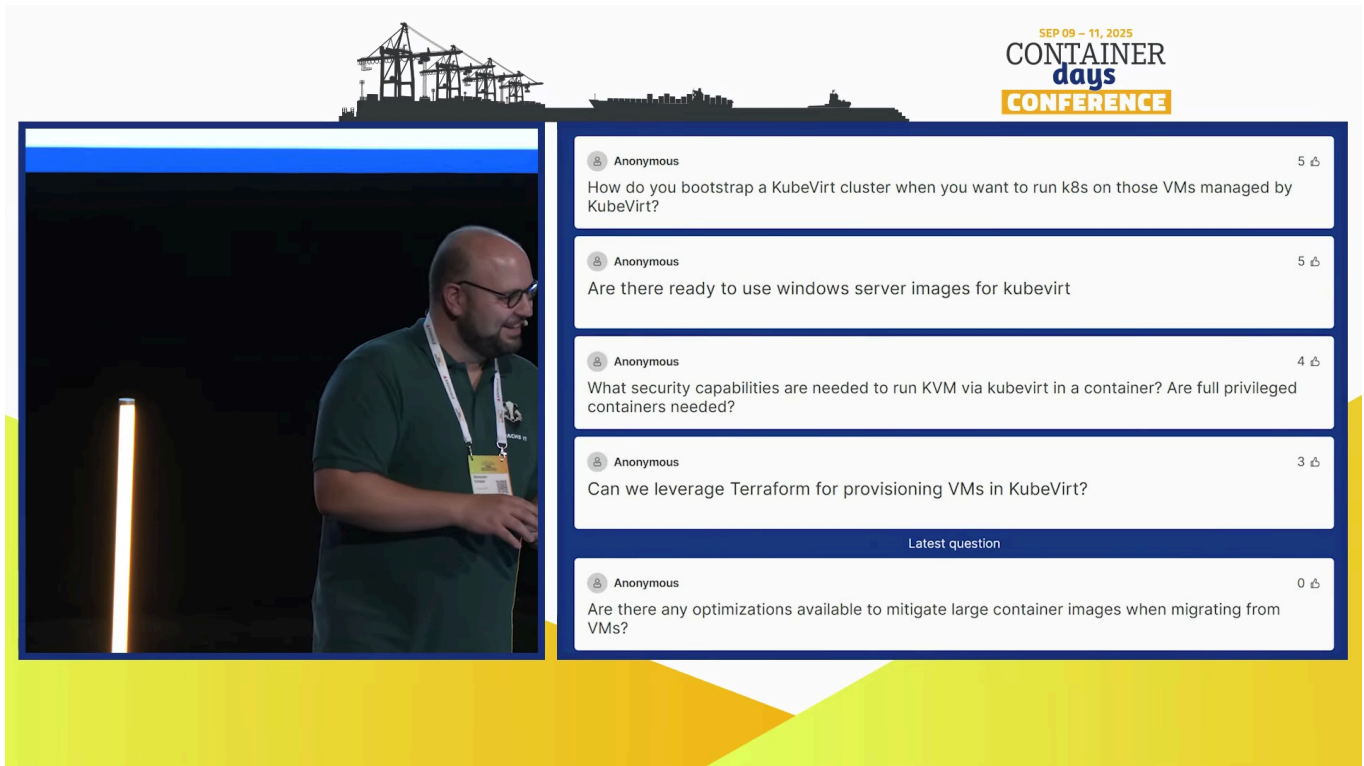
to tackle this topic. Well, we kind of said it's a transformation journey, right? And we wanted to give you all the basics. So when you start your journey, you have, you know, the terminology down, you know what they were talking about, those different tools. So I hope we got that. And now let's talk a little bit about what to actually do. So it's easy, right? It's just right the animals, right? So we have just seen it. We just have to use it now and write it. But of course, it's just half of the story. One of the bigger stories obviously is a data itself, right? Because at the end of the day, you say, the animals are great for, you know, defining your state of the virtual machines, whatever you need. But it only is really useful if you have your correct data for that. So data is obviously the operating system itself or even even full VM disks, which contain all the data, the image as your own. And so you want to kind of leverage that to make a VM which actually does something. I actually saw an example that utilize the CDI to say, listen, I want to have the Fedora Cloud image, which is

Slide 62 (at 2099.0s)

 **Timestamp:** 2099.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=2099>

 **Slide Image**



The slide image shows a presentation screen at the Container Days Conference. On the left, a man in a green shirt is speaking. The screen displays a list of questions from the audience, all submitted by 'Anonymous' users. The questions are related to KubeVirt, Windows server images, KVM security, Terraform, and container image optimizations. The conference logo 'CONTAINER days CONFERENCE' and the dates 'SEP 09 - 11, 2025' are visible in the top right corner of the screen.

SEP 09 - 11, 2025
CONTAINER days
CONFERENCE

Anonymous 5 ⬆
How do you bootstrap a KubeVirt cluster when you want to run k8s on those VMs managed by KubeVirt?

Anonymous 5 ⬆
Are there ready to use windows server images for kubevirt

Anonymous 4 ⬆
What security capabilities are needed to run KVM via kubvirt in a container? Are full privileged containers needed?

Anonymous 3 ⬆
Can we leverage Terraform for provisioning VMs in KubeVirt?

Latest question

Anonymous 0 ⬆
Are there any optimizations available to mitigate large container images when migrating from VMs?

 **Transcript**

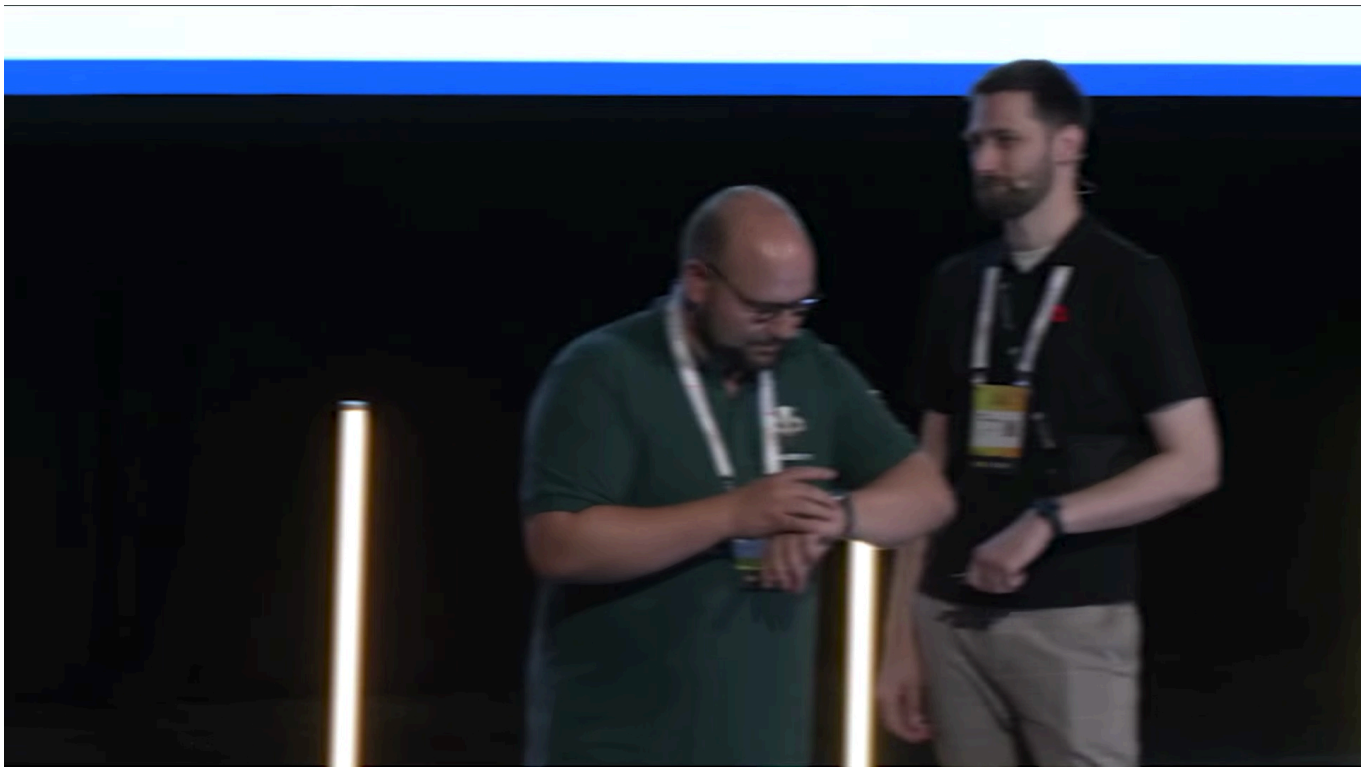
have to use it now and write it. But of course, it's just half of

Slide 63 (at 2104.0s)

 **Timestamp:** 2104.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=2104>

 **Slide Image**



Transcript

contain all the

Slide 64 (at 2105.0s)

 **Timestamp:** 2105.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=2105>

Slide Image



Transcript

the standardization of

Slide 65 (at 2106.0s)

 **Timestamp:** 2106.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=2106>

Slide Image



Transcript

kind of get

Slide 66 (at 2107.0s)

 **Timestamp:** 2107.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=2107>

Slide Image



Transcript

a process that

Slide 67 (at 2108.0s)

 **Timestamp:** 2108.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=2108>

Slide Image



Transcript

custom resources you

Slide 68 (at 2109.0s)

 **Timestamp:** 2109.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=2109>

Slide Image



Transcript

resources like, for

Slide 69 (at 2110.0s)



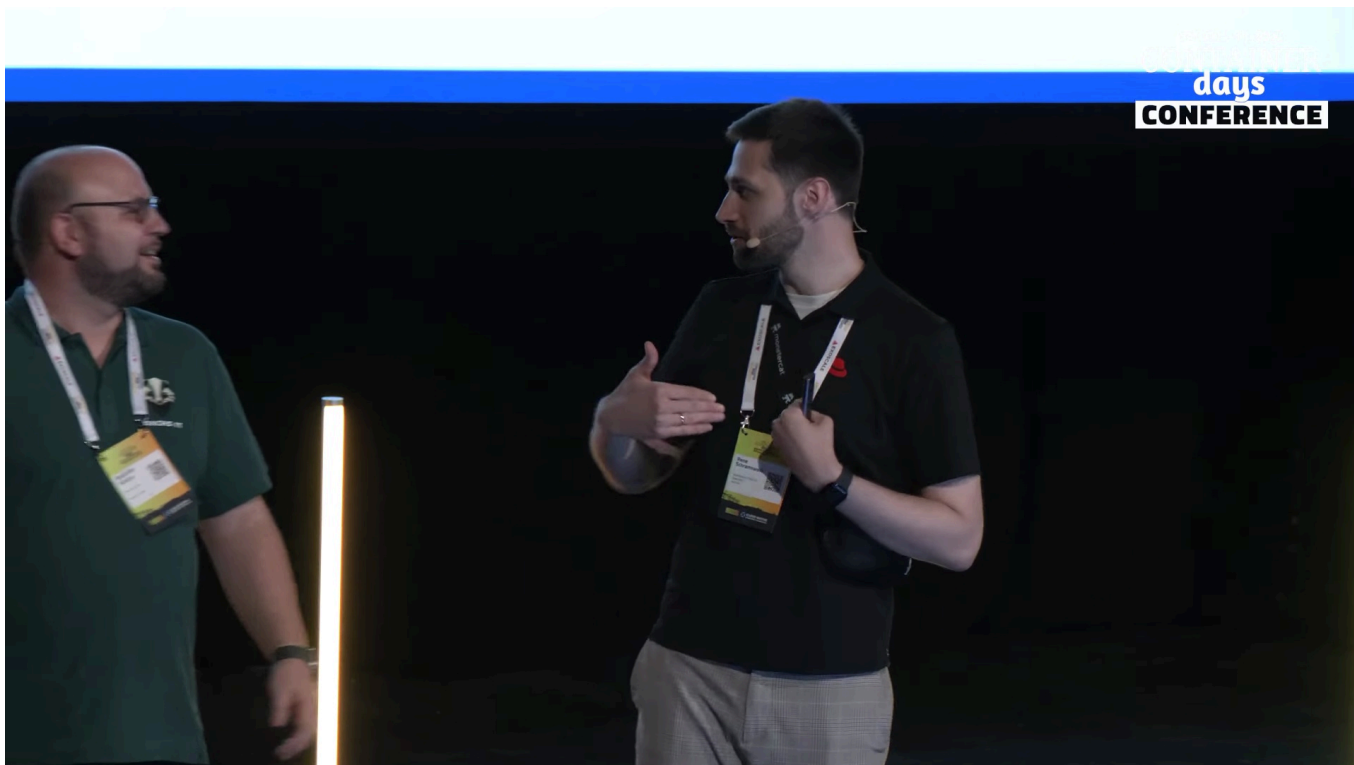
Timestamp: 2110.0s



YouTube Link: <https://www.youtube.com/watch?t=2110>



Slide Image



Transcript

to kind of

Slide 70 (at 2111.0s)

 **Timestamp:** 2111.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=2111>

Slide Image



Transcript

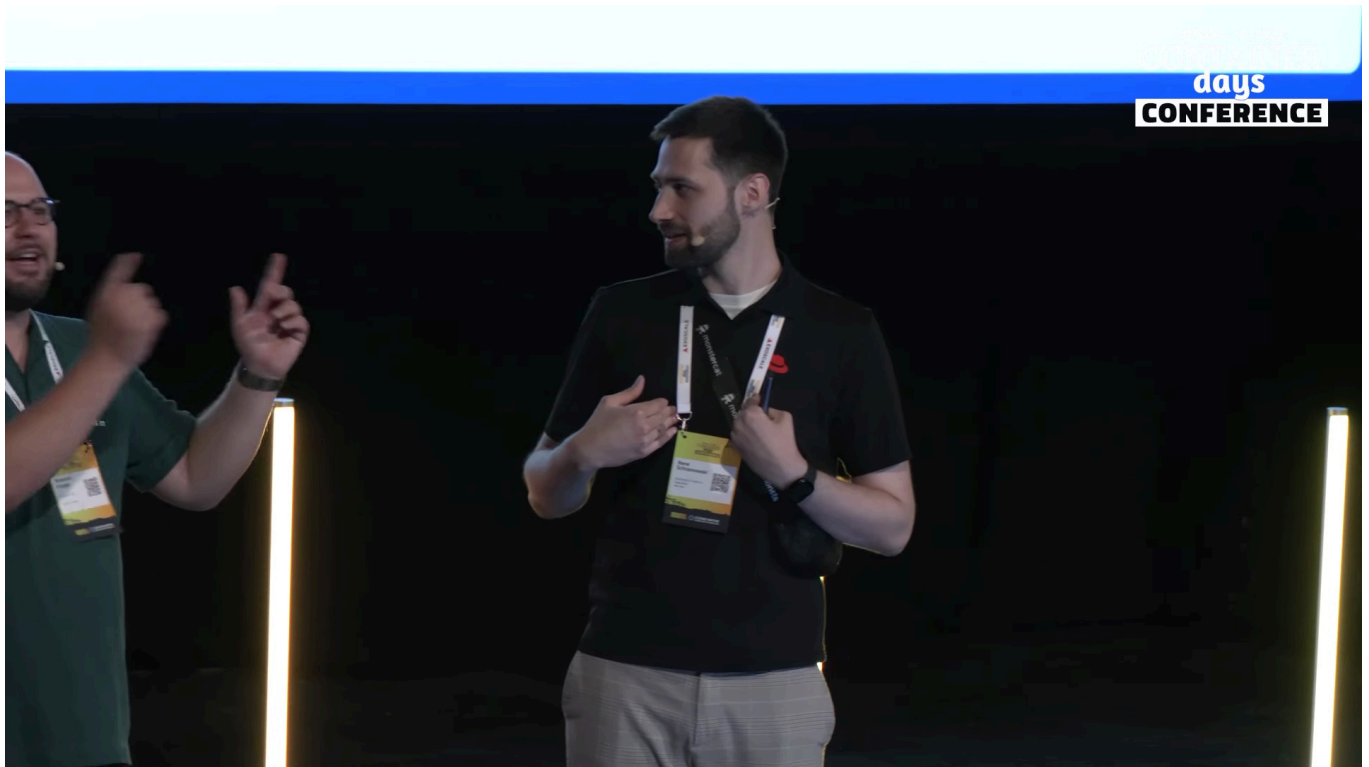
of really tells

Slide 71 (at 2112.0s)

 Timestamp: 2112.0s

 YouTube Link: <https://www.youtube.com/watch?t=2112>

Slide Image



Transcript

to look into

Slide 72 (at 2113.0s)



Timestamp: 2113.0s



YouTube Link: <https://www.youtube.com/watch?t=2113>



Slide Image



Transcript

then into your

Slide 73 (at 2114.0s)



Timestamp: 2114.0s



YouTube Link: <https://www.youtube.com/watch?t=2114>



Slide Image



Transcript

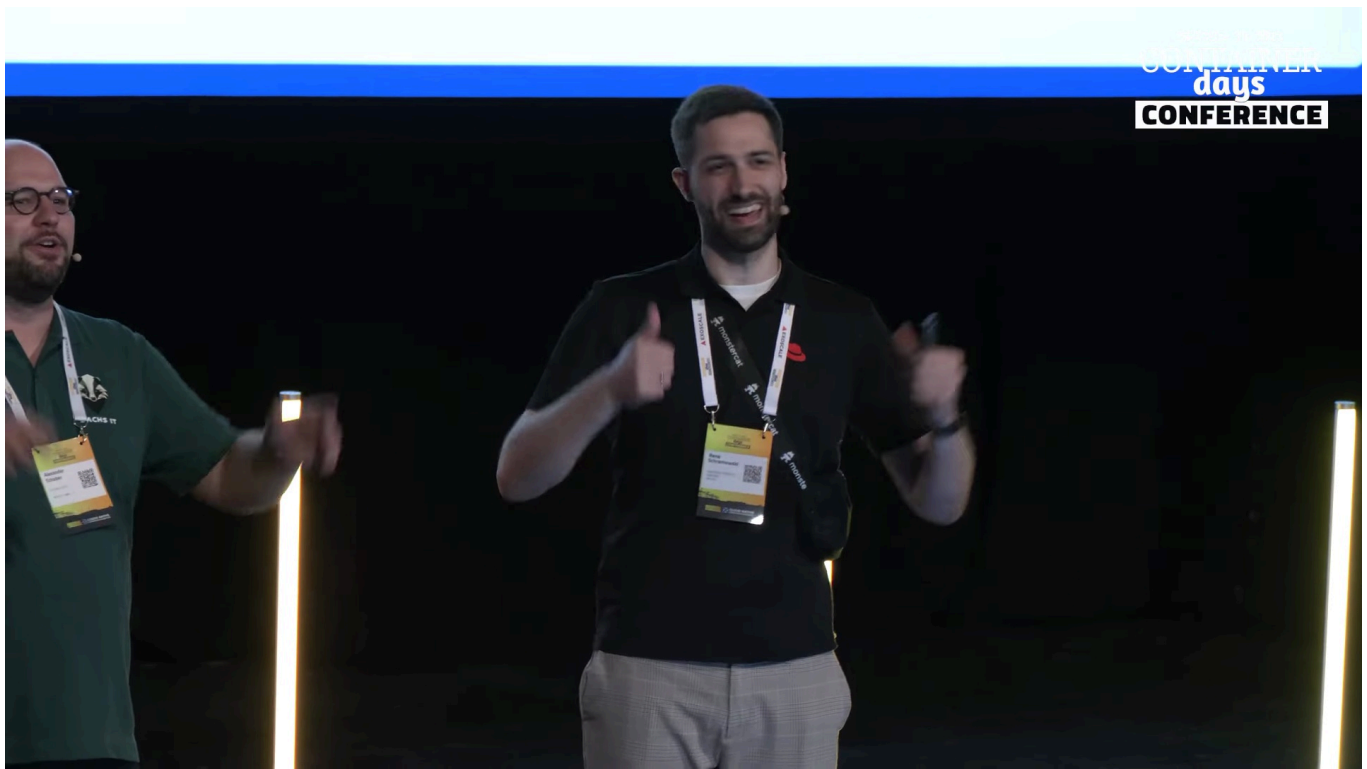
are going to

Slide 74 (at 2115.0s)

 **Timestamp:** 2115.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=2115>

Slide Image



Transcript

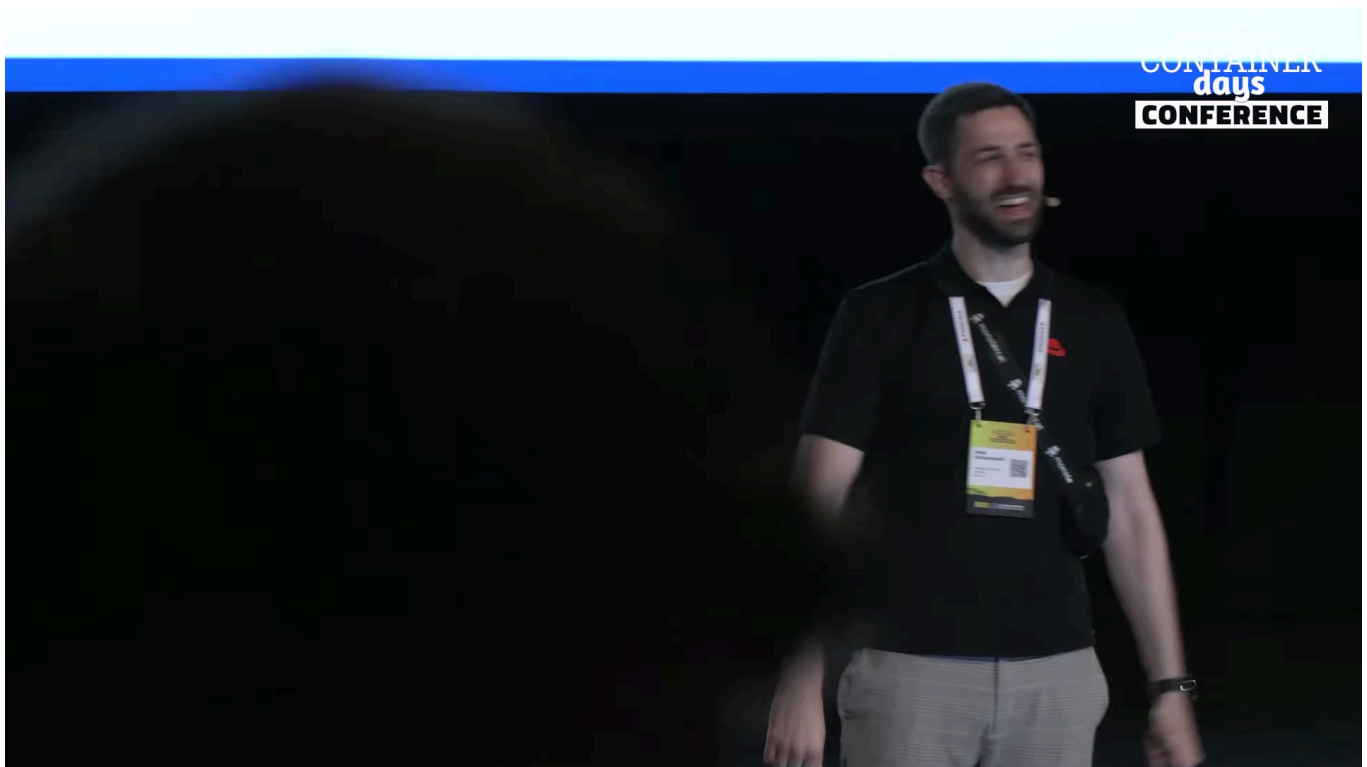
is something you

Slide 75 (at 2116.0s)

 **Timestamp:** 2116.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=2116>

Slide Image



Transcript

the virtualization knowledge

Slide 76 (at 2117.0s)

 **Timestamp:** 2117.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=2117>

Slide Image



Transcript

of the box.

Slide 77 (at 2118.0s)

 **Timestamp:** 2118.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=2118>

Slide Image



Transcript

ask about our

Slide 78 (at 2119.0s)

 **Timestamp:** 2119.0s

 **YouTube Link:** <https://www.youtube.com/watch?t=2119>

Slide Image



Transcript

bridge the gap

Slide 79 (at 2120.0s)



Timestamp: 2120.0s



YouTube Link: <https://www.youtube.com/watch?t=2120>



Slide Image



Transcript

interesting question. First of all, running containers. So we're not talking about VMs right now because you don't want to reach out. But I think

Slide 80 (at 2128.0s)



Timestamp: 2128.0s



YouTube Link: <https://www.youtube.com/watch?t=2128>



Slide Image

SEP 09 – 11, 2025

CONTAINER days CONFERENCE



Transcript

considered a invite everybody to try

Slide 81 (at 2130.0s)



Timestamp: 2130.0s



YouTube Link: <https://www.youtube.com/watch?t=2130>



Slide Image



Transcript

a limit. That's not considered safe anymore. So I don't know how much time we have. I think we're 36. Yeah, no worries. Thank you all for being here. You can ask all the questions later on. We'll be going to be here for a while. Come on by. Come on to our booth. We're right outside the door. We got a slushies for you. Thank you very much. Thank you. From container days, Hamburg, this video is proudly presented by Ducks IT.



Summary

- **Total Slides:** 81
 - **Video Source:** <https://www.youtube.com/watch?v=nShiALbl6D8>
 - **Exported:** 2025-11-03 20:28:46
 - **Generated by:** YouTube Webinar Summarizer
-

This markdown file contains embedded images as base64 data. It's compatible with most markdown viewers including GitHub, GitLab, Notion, and Obsidian.